QAOA for Portfolio Optimisation

FYS5419 – Quantum Computing
and Quantum Machine Learning

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Abstract

Picking K of n candidate stocks to balance return against risk is convex when the weights are continuous and combinatorially hard when they are binary, which makes cardinality-constrained Markowitz portfolio selection a natural test case for the Quantum Approximate Optimisation Algorithm (QAOA). We build a pure-NumPy QAOA simulator from scratch and run eight experiments on a single canonical $n = 16$, $K = 4$ instance drawn from a 16-equity universe assembled with a wink at the course title (mega-cap tech, quantum-computing pure-plays, large-cap quantum-adjacent firms, and a macro-opposed mix); a quantum algorithm should at least be allowed to pick quantum stocks. We benchmark against brute force, greedy-Sharpe, rounded Markowitz, simulated annealing, and an exact CBC mixed-integer programme.

All five classical baselines return the same four-asset basket (GOOGL, IBM, NVDA, JPM) in milliseconds. Standard penalty-mixer QAOA at depth three reaches a scaled approximation ratio of $r \approx 0.99$, but its most likely outcome is the infeasible all-ones state, with the four next ranks all single-asset selections; it has learnt the broad shape of the penalty term and not the cardinality constraint. Two known fixes close most of the gap: warm-started angles restore monotone r-versus-depth scaling, and a constraint-preserving XY-ring mixer with a Dicke initial state drives feasibility to one by construction and lifts r to 0.9995. No quantum advantage is claimed; the contribution is a transparent characterisation of where penalty QAOA fails on this problem and a clean empirical confirmation of two structural fixes from the literature.

1 Introduction

Modern portfolio theory begins with the mean–variance framework of Markowitz (1952), in which an investor trades expected return against the variance of a weighted basket of assets. In plain words, a portfolio is a collection of stocks held in chosen proportions, and the goal is to earn the most return for a given amount of risk. The expected portfolio return $\mu_p = \sum_i w_i \mu_i$ is the weighted average of asset means, and the portfolio variance $\sigma_p^2 = \sum_{ij} w_i w_j \Sigma_{ij}$ collects the cross-asset correlations into a single number; their ratio μ_p/σ_p is the *Sharpe ratio*, the industry's headline reward-per-unit-risk benchmark, and the Markowitz trade-off writes the same idea explicitly as $\mu_p - \lambda\sigma_p^2$, with $\lambda > 0$ a risk-aversion parameter. Figure 1 shows the sixteen daily price series we work with; fig. 2 places each ticker on the risk-return plane that any portfolio choice has to navigate. The continuous problem is a convex quadratic programme with a closed-

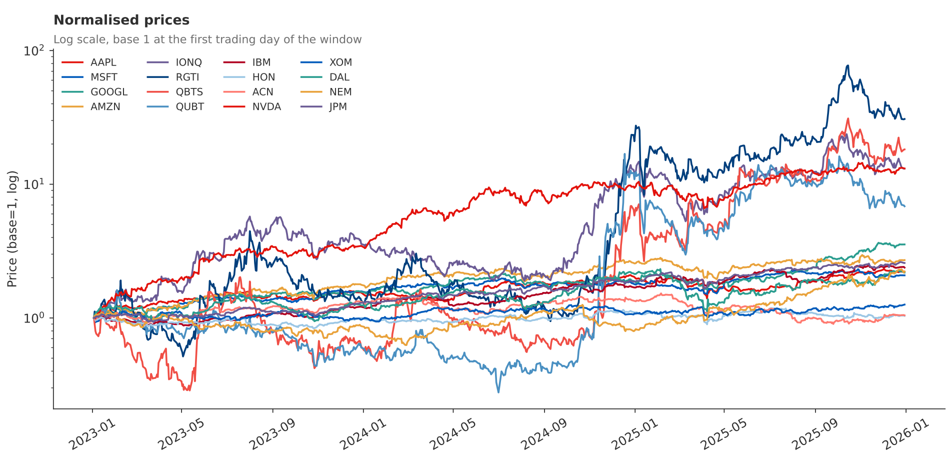

Figure 1. The sixteen-equity investable universe used throughout this report, with prices normalised to unity at the start of the sample (2023-01-03 to 2025-12-30, 751 trading days). The four small-cap quantum-computing names (IONQ, RGTI, QBTS, QUBT) trace the upper envelope with order-of-magnitude swings; the mega-cap tech and quantum-adjacent baskets follow a steadier upward trajectory. The task of this report is to pick a fixed-size subset of these sixteen series whose combined holding balances expected return against return variance.

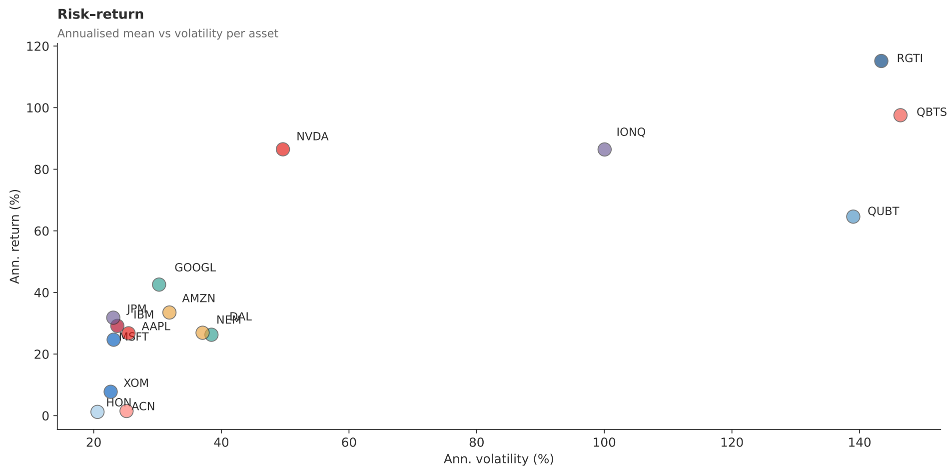

Figure 2. Annualised risk-return scatter for the sixteen-equity universe (751 trading days, 2023-01-03 to 2025-12-30). Horizontal axis: annualised volatility $\sqrt{252}\,\sigma(r_i)$; vertical axis: annualised mean return $252\,\langle r_i \rangle$; markers coloured by basket. NVDA sits at the top of the cloud with the highest single-asset Sharpe ratio (1.74); JPM and GOOGL form the next-best risk-adjusted cluster, while the four small-cap quantum names sit on the high-volatility right edge. Any portfolio choice combines these points into a single weighted location on the same plane, and the cardinality-constrained Markowitz problem of this report asks which K-asset subset lands as far to the upper-left as possible.

form solution. Practical mandates almost always overlay discrete structure: a fund may have to hold exactly K of n candidate equities, allocate in integer lots, or respect sector caps. The cardinality constraint $\sum_i x_i = K$ on binary selectors $\boldsymbol{x} \in \{0,1\}^n$ collapses the convex programme onto a combinatorial lattice of size $\binom{n}{K}$ and turns mean–variance optimisation into an NP-hard binary quadratic programme. Brute-force enumeration scales as 2^n. At our $n = 16$ instance the search space is $2^{16} \approx 65{,}536$ bitstrings, enumerable in 30 ms on a laptop. The same Quadratic Unconstrained Binary Optimisation (QUBO) form recurs across MAX-CUT, the travelling salesman problem, and discrete knapsack-style allocation (Huot et al. 2024), so the methods reach well beyond finance.

The Quantum Approximate Optimisation Algorithm (QAOA), introduced by Farhi et al. (2014), recasts QUBO minimisation as a ground-state preparation problem for an Ising Hamiltonian H_C. A variational ansatz alternates p layers of a cost unitary $e^{-i\gamma_k H_C}$ with a mixing unitary $e^{-i\beta_k H_M}$ acting on a uniform superposition, and the $2p$ angles $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ are tuned by a classical outer loop. The Trotterised ansatz recovers the adiabatic algorithm in the limit $p \to \infty$ (Zhou et al. 2018). With n qubits, depth $O(p)$ and only $2p$ parameters, QAOA is a flagship Noisy Intermediate-Scale Quantum (NISQ) algorithm and a cousin of the variational quantum eigensolver (Tilly et al. 2022; Giovagnoli 2020).

Portfolio selection has become a flagship QAOA use case. Stopfer et al. benchmark depth and noise across realistic asset universes (Stopfer and Wagner 2025); Mancilla et al. study a constraint-preserving XY-mixer that keeps the state in the cardinality-feasible subspace (Mancilla et al. 2026); Thomassin et al. propose a slack-variable encoding for inequality constraints (Thomassin et al. 2025); Shen et al. quantify the gap between idealised and noisy-device performance (Shen et al. 2025); Morapakula et al. compare QAOA against quantum annealing (Morapakula et al. 2025); Soloviev et al. reduce measurement overhead via operator pools (Soloviev and Krompiec 2025) and push simulators past $n = 16$ with circuit cutting (Soloviev, Márquez Romero, et al. 2025). The conclusion is consistent: QAOA probes QUBO landscapes informatively but does not yet rival classical solvers at practical scale.

Honesty about QAOA's limits matters. No quantum advantage has been demonstrated for QAOA on any useful problem, portfolio selection included. Classical baselines comfortably outperform QAOA at the depths achievable today. The optimisation step is plagued by *barren plateaus*: as the problem grows the cost landscape becomes exponentially flat for random initialisations of sufficiently deep circuits, so a gradient-based or random search has essentially nothing to follow towards the optimum (McClean et al. 2018). The depth p required to compete with classical heuristics on hard instances appears to grow with system size, eroding the constant-depth NISQ picture. At our $n = 16$ instance, brute force returns the certified optimum in tens of milliseconds and CBC (Forrest and Lougee-Heimer 2005) closes it to proven optimality almost as quickly; a single QAOA run at $p = 3$ with fifty COBYLA restarts takes minutes, several orders of magnitude slower for an answer that concentrates on the wrong bitstrings. We therefore treat QAOA as a research vehicle for studying cost-landscape geometry, depth scaling and the interference mechanism. It is not meant to beat CBC or Gurobi on a 16-equity Markowitz instance.

The work reported here is methodological and built around a single sharp finding together with two structural fixes that almost close the gap. We implement QAOA from scratch in pure NumPy, building the statevector simulator, the Ising-mapped cost Hamiltonian and the COBYLA outer loop without recourse to Qiskit (IBM Quantum 2026) or PennyLane (Xanadu 2026). The pipeline runs end to end: daily equity prices from 2023-01-03 to 2025-12-30 (751 observations on a 16-asset universe drawn from the mag7, quantum, quantum-big and anti baskets) yield moments $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, folded into a penalised QUBO with cardinality term $A(\sum_i x_i - K)^2$, mapped to an Ising Hamiltonian and realised as a Pauli-Z cost operator H_C paired with a transverse-field mixer H_M. Every run is validated against brute force and, at the canonical instance, against a CBC MIP solve. The instance falsifies the standard recipe. Four classical baselines (exhaustive enumeration, greedy-Sharpe selection, rounded continuous Markowitz and an exact CBC MIP, with simulated annealing as a fifth check) all converge in milliseconds to the same basket, $\{\mathrm{GOOGL}, \mathrm{IBM}, \mathrm{NVDA}, \mathrm{JPM}\}$. Standard QAOA at $p = 3$ reaches a scaled approximation ratio $r \approx 0.99$ with fifty restarts, which sounds respectable. Yet the top-1 most probable bitstring is the all-ones state "pick every asset", and ranks two through five are single-asset quantum stocks. The algorithm gets confused about what "pick four" means and votes instead for "pick sixteen" followed by "pick one"; probability on the true optimum is roughly an order of magnitude above the uniform baseline. Eight experiments anatomise this failure. A canonical single-run benchmark locates QAOA on the cost spectrum. A full $p = 1$ energy landscape over (γ, β) visualises the basins and saddles the outer loop must navigate. Four sweeps vary one knob at a time (depth p, problem size n, risk aversion λ and budget penalty A) to characterise when the interference mechanism breaks down. A warm-start ablation in the spirit of Zhou et al. (2018) separates restart-budget pathology from genuine ansatz limits and restores monotone r-versus-p scaling. A head-to-head with a Hamming-weight-preserving XY-ring mixer paired with a Dicke initial state (Mancilla et al. 2026; Bärtschi and Eidenbenz 2019) shows that exact feasibility raises the approximation ratio to near-unity at every depth tested, at an order-of-magnitude per-layer simulator cost.

The remainder is organised as follows. Section 2 develops the formalism from the Markowitz objective through the QUBO and Ising mappings to the QAOA circuit, the XY-mixer variant and the warm-start prescription. Section 3 describes the pure-NumPy implementation, the equity dataset, the MIP and XY-mixer modules and the experimental protocol. Section 4 reports the eight experiments alongside the brute-force, greedy, Markowitz-round, simulated-annealing and MIP baselines. Section 5 summarises the findings, draws out the lessons of the warm-start and XY-mixer ablations and revisits the caveats above.

2 Methods

The QAOA pipeline turns a financial dataset into a quantum circuit through a sequence of well-defined steps,

$$\text{Financial data} \to (\boldsymbol{\mu}, \boldsymbol{\Sigma}) \to \text{QUBO} \to \text{Ising} \to \text{Pauli operators} \to \text{QAOA}. \tag{1}$$

In plain words: we start from daily stock prices, summarise them as an expected-return vector $\boldsymbol{\mu}$ and a covariance matrix $\boldsymbol{\Sigma}$, write the portfolio objective as a quadratic cost over binary "in or out" decisions (a QUBO), rewrite the same cost in ± 1-spin variables so it becomes a classical magnet (an Ising Hamiltonian), promote each spin to a Pauli-Z operator on a qubit, and finally drive a short tunable quantum circuit (QAOA) whose measurement statistics concentrate on low-cost portfolios. Subsections 2.1–2.5 develop each arrow in turn. Section 2.6 develops the constraint-preserving XY-ring mixer that we later benchmark in section 4.8, section 2.7 formalises the warm-start initialisation prescription benchmarked in section 4.4, and section 2.8 fixes the metrics used throughout section 4.

2.1 *Mean–variance portfolio optimisation*

For a binary selection vector $\boldsymbol{x} \in \{0, 1\}^n$ over n candidate assets, the Markowitz objective (Markowitz 1952) is

$$\min_{\boldsymbol{x}} \left(-\boldsymbol{\mu}^\top \boldsymbol{x} + \lambda\, \boldsymbol{x}^\top \boldsymbol{\Sigma}\, \boldsymbol{x} \right), \tag{2}$$

where $-\boldsymbol{\mu}^\top \boldsymbol{x}$ maximises expected return and $\lambda\, \boldsymbol{x}^\top \boldsymbol{\Sigma}\, \boldsymbol{x}$ penalises portfolio variance. The risk-aversion parameter $\lambda \geq 0$ tunes the trade-off.

A fixed-budget constraint requires exactly K assets,

$$\sum_{i=1}^{n} x_i = K, \tag{3}$$

which we enforce with a quadratic penalty, giving the soft-constrained cost

$$C(\boldsymbol{x}) = -\boldsymbol{\mu}^\top \boldsymbol{x} + \lambda\, \boldsymbol{x}^\top \boldsymbol{\Sigma}\, \boldsymbol{x} + A \left(\sum_{i=1}^{n} x_i - K \right)^2, \tag{4}$$

where $A > 0$ is chosen large enough that the global minimum of $C(\boldsymbol{x})$ satisfies the budget constraint. Section 4.5 studies the sensitivity of this calibration empirically.

Returns and their first two moments are estimated from price series $P_i(t)$ via log returns,

$$r_i(t) \;=\; \ln \frac{P_i(t+1)}{P_i(t)}, \tag{5}$$

$$\mu_i = \langle r_i \rangle_t, \qquad \Sigma_{ij} = \langle r_i r_j \rangle_t - \langle r_i \rangle_t \langle r_j \rangle_t, \tag{6}$$

where $\langle \cdot \rangle_t$ denotes the time average across the in-sample window. Estimating $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ from a single window is a well-known weakness of the Markowitz programme (Hastie et al. 2009); we accept it here, since the algorithmic question is independent of the estimator.

2.2 *Quadratic unconstrained binary optimisation (QUBO)*

QUBO (Quadratic Unconstrained Binary Optimisation) is the standard input format the quantum optimisation literature expects, so the penalised mean–variance cost of eq. (4) is rewritten in that shape before anything quantum happens. A QUBO is a quadratic cost over binary variables,

$$\min_{\boldsymbol{x}\in\{0,1\}^n} C(\boldsymbol{x}) = \sum_i Q_{ii}\, x_i + \sum_{i<j} Q_{ij}\, x_i x_j, \tag{7}$$

where the linear terms are absorbed into the diagonal of Q because $x_i^2 = x_i$ for binary variables. Expanding the penalised mean–variance cost (4) yields

$$\begin{aligned} C(\boldsymbol{x}) = \sum_i \underbrace{\left[-\mu_i + \lambda\Sigma_{ii} + A(1-2K)\right]}_{Q_{ii}} x_i \\ + \sum_{i<j} \underbrace{\left[2\lambda\Sigma_{ij} + 2A\right]}_{Q_{ij}} x_i x_j + \underbrace{AK^2}_{\text{const}}, \end{aligned} \tag{8}$$

so

$$\boxed{\begin{aligned} Q_{ii} &= -\mu_i + \lambda\Sigma_{ii} + A(1-2K), \\ Q_{ij} &= 2\lambda\Sigma_{ij} + 2A \quad (i<j). \end{aligned}} \tag{9}$$

2.3 *Ising Hamiltonian mapping*

The next step rewrites the same cost in ± 1-spin variables rather than $\{0,1\}$ bits. The reason is purely book-keeping for the quantum step that follows: the Pauli-Z operator that will replace each spin has single-qubit eigenvalues ± 1 by construction, so spins map to qubits with no extra arithmetic. Mapping binary variables to spins via

$$x_i = \tfrac{1}{2}(1 - z_i), \qquad z_i \in \{-1, +1\}, \tag{10}$$

and substituting into $C(\boldsymbol{x})$ produces an Ising form

$$H(\boldsymbol{z}) = \sum_i h_i\, z_i + \sum_{i<j} J_{ij}\, z_i z_j, \tag{11}$$

with coefficients

$$h_i = -\tfrac{1}{2} Q_{ii} - \tfrac{1}{4} \sum_{j\neq i} Q_{ij}, \qquad J_{ij} = \tfrac{1}{4} Q_{ij} \quad (i<j), \tag{12}$$

and an additive constant

$$c = AK^2 + \tfrac{1}{2} \sum_i Q_{ii} + \tfrac{1}{4} \sum_{i<j} Q_{ij}, \tag{13}$$

chosen so that $H(\boldsymbol{z}) + c = C(\boldsymbol{x})$ for every binary $\boldsymbol{x}$ and its spin image $\boldsymbol{z}$. We keep the constant in the Hamiltonian throughout this report, so the eigenvalues of H_C defined below equal the cost values $C(\boldsymbol{x})$ directly, the same convention as the reference implementation.

2.4 *Pauli operators and the cost Hamiltonian*

We follow the standard quantum-computing conventions of Nielsen and Chuang (2000) for single-qubit gates and tensor-product ordering. Promoting each classical spin z_i to the single-qubit Pauli-Z operator on site i,

$$z_i \longrightarrow \hat{Z}_i = I^{\otimes i} \otimes Z \otimes I^{\otimes(n-i-1)}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{14}$$

gives the cost Hamiltonian as the $2^n \times 2^n$ diagonal matrix

$$\boxed{H_C = c\, I + \sum_i h_i\, \hat{Z}_i + \sum_{i<j} J_{ij}\, \hat{Z}_i \hat{Z}_j.} \tag{15}$$

Every computational basis state $|\boldsymbol{z}\rangle$ is an eigenstate of H_C with eigenvalue $C(\boldsymbol{x})$.

The standard *transverse-field* mixing Hamiltonian drives tunnelling between bitstring states,

$$H_M = \sum_{i=1}^{n} \hat{X}_i, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{16}$$

Two properties make H_M natural. First, $\hat{X}_i$ flips qubit i, so $[H_M, H_C] \neq 0$ and the mixer generates the non-diagonal dynamics QAOA needs to interfere between bitstrings. Second, the ground state of $-H_M$ is the uniform superposition

$$|s\rangle \;=\; |+\rangle^{\otimes n} \;=\; \frac{1}{\sqrt{2^n}} \sum_{\boldsymbol{x}\in\{0,1\}^n} |\boldsymbol{x}\rangle, \tag{17}$$

prepared by a layer of Hadamards on $|0\rangle^{\otimes n}$ and used as the initial state of the variational circuit.

2.5 *QAOA ansatz*

In one sentence, QAOA is a short tunable quantum circuit whose angles are chosen so that the most likely measurement outcomes are the lowest-cost portfolios. The remainder of this subsection unpacks what *short*, *tunable*, and *most likely* mean in practice.

Adiabatic intuition. QAOA (Farhi et al. 2014) is a discretised, variational analogue of quantum adiabatic computation. The adiabatic theorem says that a system prepared in the ground state of an initial Hamiltonian and slowly evolved towards a final Hamiltonian stays in the instantaneous ground state. Taking H_C from section 2.4 as the final Hamiltonian and the transverse-field mixer H_M as the initial Hamiltonian, with ground state (17), turns combinatorial optimisation into ground-state preparation. The required evolution time can scale exponentially in n, so QAOA approximates it by p short alternating steps generated by H_C and H_M and treats the step durations (γ_k, β_k) as variational parameters. As $p \to \infty$ the ansatz recovers the exact adiabatic algorithm (Zhou et al. 2018).

The single-layer ($p = 1$) ansatz is

$$|\psi(\gamma, \beta)\rangle = e^{-i\beta H_M}\, e^{-i\gamma H_C}\, |+\rangle^{\otimes n}, \tag{18}$$

where $e^{-i\gamma H_C}$ is the *phase-separation* unitary (it phases each basis state by $e^{-i\gamma\, C(\boldsymbol{x})}$, encoding cost information into the amplitudes), and $e^{-i\beta H_M}$ is the *mixing* unitary, which redistributes amplitude between bitstrings to enable transitions between candidate solutions.

$$\underbrace{H^{\otimes n}}_{\text{superposition}} \longrightarrow \underbrace{e^{-i\gamma H_C}}_{\text{phase separation}} \longrightarrow \underbrace{e^{-i\beta H_M}}_{\text{mixing}} \longrightarrow \underbrace{\text{measure}}_{Z\text{-basis}} \tag{19}$$

For depth $p > 1$, p cost and p mixer layers are alternated,

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} \left(e^{-i\beta_k H_M}\, e^{-i\gamma_k H_C}\right) |+\rangle^{\otimes n}, \qquad (20)$$

with $2p$ variational parameters $(\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p)$.

Interference mechanism. The diagonal $e^{-i\gamma H_C}$ imprints the cost landscape onto the wavefunction without changing measurement probabilities. The non-diagonal $e^{-i\beta H_M}$ mixes amplitude between basis states; because $[H_C, H_M] \neq 0$, the two operations together produce nontrivial quantum interference. With the right angles, amplitude accumulates constructively on low-cost bitstrings and destructively on high-cost ones, concentrating measurement probability on near-optimal solutions. This is what distinguishes QAOA from classical heuristics such as simulated annealing or tabu search: amplitude flows non-locally across the cost landscape in superposition, whereas classical samplers visit one configuration at a time (Giovagnoli 2020).

Hybrid quantum–classical loop. A classical optimiser minimises the energy expectation value

$$E(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})|\, H_C\, |\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle. \qquad (21)$$

Since H_C is diagonal in the computational basis with eigenvalues $C(\boldsymbol{x})$, this reduces to

$$E(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \sum_{\boldsymbol{x}\in\{0,1\}^n} C(\boldsymbol{x})\, |\langle\boldsymbol{x}|\psi\rangle|^2, \qquad (22)$$

so the QAOA energy is a probability-weighted average of cost values. The circuit has $2p$ parameters and depth $O(p)$: the quantum device supplies a non-classical sampling primitive, while the parameter search sits on a classical CPU (Tilly et al. 2022). The variational principle guarantees $E(\boldsymbol{\gamma}, \boldsymbol{\beta}) \geq E_{\text{opt}}$ for all angles, with equality as $p \to \infty$. Our classical optimiser is COBYLA with random multistarts; see section 3.1.

Optimiser pathologies. The classical outer loop is the dominant practical bottleneck. Two failure modes are known. First, *barren plateaus*: for sufficiently deep random parameterised circuits the variance of $\partial E/\partial\theta$ vanishes exponentially in n, so gradient-based or random-search methods need exponentially many samples to make progress (McClean et al. 2018). The transverse-field mixer ansatz studied here is, formally, in the regime where this prediction applies once p is large; the warm-start prescription of section 2.7 is one of the standard mitigations. Second, even outside the barren-plateau regime the angle space $(\boldsymbol{\gamma}, \boldsymbol{\beta}) \in \mathbb{R}^{2p}$ is non-convex, and a fixed random-multi-start budget covers it exponentially less densely as p grows. Sections 4.3 and 4.4 report both effects empirically.

2.6 XY-mixer: a constraint-preserving alternative

The XY-mixer answers a simple question: instead of *penalising* portfolios that pick the wrong number of assets, what if the algorithm physically *cannot leave* the subspace of valid K-asset portfolios in the first place? Then the penalty in (4) can be set to zero. The penalty $A(\sum_i x_i - K)^2$ in (4) enforces the budget only as $A \to \infty$. At finite A, infeasible bitstrings remain in the search space and may carry non-negligible amplitude after a finite-depth circuit. An alternative drops the penalty and uses a mixer that preserves Hamming weight by construction (Mancilla et al. 2026). The *XY-ring mixer* is

$$H_M^{(XY)} = \tfrac{1}{2}\sum_{i=1}^{n}\big(\hat{X}_i\hat{X}_{i+1} + \hat{Y}_i\hat{Y}_{i+1}\big), \qquad (23)$$

with periodic boundary conditions ($i+1$ taken mod n) and

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \qquad (24)$$

The operator $\hat{X}_i\hat{X}_{i+1} + \hat{Y}_i\hat{Y}_{i+1}$ swaps $|01\rangle$ and $|10\rangle$ on neighbouring qubits while leaving $|00\rangle$ and $|11\rangle$ invariant, so it commutes with the Hamming-weight operator $\hat{N} = \sum_i \frac{1}{2}(I - \hat{Z}_i)$. If the initial state is a feasible Dicke state $|D_K^n\rangle$, the entire QAOA trajectory stays in the feasible subspace and the penalty A becomes unnecessary.

The XY-mixer is hyperparameter-free but requires Dicke-state preparation and has more gates per layer than (16). In our statevector simulator the per-layer cost is dominated by a sparse `expm_multiply` on a $2^n \times 2^n$ matrix, whereas the transverse-field mixer factorises into n single-qubit rotations and is roughly an order of magnitude cheaper. The Dicke state $|D_K^n\rangle$, the equal superposition over all weight-K bitstrings,

$$|D_K^n\rangle \;=\; \frac{1}{\sqrt{\binom{n}{K}}} \sum_{\boldsymbol{x}:\,|\boldsymbol{x}|=K} |\boldsymbol{x}\rangle, \qquad (25)$$

is the symmetric ground state of $-H_M^{(XY)}$ in that subspace, and the preparation circuit of Bärtschi and Eidenbenz (2019) runs in depth $O(n)$ on a fully connected device. Section 4.8 benchmarks XY/Dicke against penalty/transverse-field at fixed n, K and depth.

2.7 Warm-started angle initialisation

The variational landscape $E(\boldsymbol{\gamma}, \boldsymbol{\beta})$ lives in $2p$ real dimensions and is non-convex even at $p = 1$ (fig. 6). Drawing all $2p$ angles uniformly from $\mathcal{U}[0, 2\pi]^{2p}$ at each depth treats the sweep as p independent problems, so a fixed restart budget covers angle space exponentially less densely as p grows. Zhou et al. (2018) noted that the optimal angles at depth p are typically continuously related to those at $p-1$, and proposed seeding the depth-p optimiser by appending a fresh layer to the depth-$(p-1)$ optimum,

$$(\boldsymbol{\gamma}_0^{(p)}, \boldsymbol{\beta}_0^{(p)}) \;=\; \big((\boldsymbol{\gamma}^{\star,(p-1)}, \gamma_p^{\text{rand}}),\, (\boldsymbol{\beta}^{\star,(p-1)}, \beta_p^{\text{rand}})\big), \qquad (26)$$

followed by a short COBYLA refinement. Iterating from $p = 1$ upwards turns the depth sweep into a single trajectory through cumulative $\mathbb{R}^{2p}$ angle space instead of p independent ones. Any non-monotonicity in r versus p that comes from angle-space coverage (rather than ansatz limits) should then disappear, the empirical test of section 4.4. The prescription adds one layer per depth at no extra restart cost and is independent of the mixer, so it stacks with the XY-mixer of section 2.6.

2.8 Metrics and validation

We track four diagnostics rather than a single number because no one of them captures both *how close the expected energy is to the optimum* and *how likely a measurement is to actually return the optimal bitstring*. The first two below are energy-based, the next two are probability-based, and we add the practitioner's Sharpe ratio so the QAOA output can be compared against the brute-force optimum on the metric an investor actually cares about.

Scaled approximation ratio. Let $E_{\text{opt}} = \min_{\boldsymbol{x}} C(\boldsymbol{x})$ and $E_{\text{worst}} = \max_{\boldsymbol{x}} C(\boldsymbol{x})$ denote the global minimum and maximum of the cost function, obtained by exhaustive enumeration. Following Farhi et al. (2014), we define

$$r \;=\; \frac{E_{\text{worst}} - E(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)}{E_{\text{worst}} - E_{\text{opt}}} \;\in\; [0,\, 1], \tag{27}$$

where $(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)$ are the COBYLA-optimised angles. $r = 1$ means the optimum is recovered in expectation; an unbiased uniform sample gives $r = (E_{\text{worst}} - \overline{E})/(E_{\text{worst}} - E_{\text{opt}})$, around 0.5 for the canonical instance at $A = 0.5$.

Relative gap. E_{worst} is dominated by the budget penalty AK^2 and inflates the denominator of (27), so the scaled ratio compresses what can be a substantial absolute shortfall (Zhou et al. 2018). We also report the unscaled relative gap

$$\text{gap}_{\text{rel}} \;=\; \frac{E(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star) - E_{\text{opt}}}{|E_{\text{opt}}|}, \tag{28}$$

which is zero at the optimum and unbounded above.

Top-1 success probability. The probability that a single measurement returns the true optimum,

$$p_{\text{opt}} \;=\; \left|\langle \boldsymbol{x}^\star | \psi(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)\rangle\right|^2, \tag{29}$$

with $\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} C(\boldsymbol{x})$ the brute-force optimum.

Feasibility probability. The probability that a measurement yields a budget-feasible bitstring,

$$p_{\text{feas}} \;=\; \sum_{\boldsymbol{x}\,:\,\sum_i x_i = K} \left|\langle \boldsymbol{x} | \psi(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)\rangle\right|^2, \tag{30}$$

which is unity by construction under the XY-mixer of section 2.6 and is a non-trivial diagnostic for the penalty mixer used here.

Sharpe ratio. For comparing candidate portfolios in financial terms alongside the QUBO cost, we report the equal-weighted Sharpe ratio of a selected basket $\boldsymbol{x}$ over the in-sample window,

$$\text{Sharpe}(\boldsymbol{x}) \;=\; \frac{\boldsymbol{\mu}^\top \boldsymbol{x}}{\sqrt{\boldsymbol{x}^\top \boldsymbol{\Sigma}\, \boldsymbol{x}}}, \tag{31}$$

computed from the same daily-return moments. This is the metric a portfolio manager would actually care about; the QUBO cost $C(\boldsymbol{x})$ of (4) is a λ-weighted proxy for it plus the budget penalty.

Validation strategy. At $n = 16$ the brute-force optimum is computable in $2^{16} \approx 6.5\times10^4$ evaluations (~ 30 ms in NumPy), so every QAOA run in section 4 is anchored against the exact answer. The QUBO→Ising mapping of (12)–(13) is also verified numerically: $H(\boldsymbol{z}) + c = C(\boldsymbol{x})$ to floating-point precision for every $|\boldsymbol{z}\rangle$, and the diagonal of H_C in (15) matches the brute-force cost vector. Further checks are in section 3.4.

3 Implementation and Datasets

3.1 Code structure

The implementation is a small collection of pure-NumPy / SciPy modules, written from scratch rather than on top of a high-level VQA library so that every conventional choice (sign of the cost, qubit ordering, scaling of the approximation ratio) is explicit and auditable. The package layout mirrors the pipeline of section 2.

Modules. The *data layer* (`data.py`) loads cached daily-close Parquet files and assembles $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The *problem layer* (`portfolio.py`, `ising.py`) fixes the canonical cost $C(\boldsymbol{x})$ via `PortfolioProblem` and `DEFAULTS`, and performs the $x_i = (1 - z_i)/2$ substitution to build the diagonal H_C and the sparse mixer H_M. The *solver layer* bundles QAOA variants (`qaoa.py`'s `solve` and `solve_warm` for the X-mixer, `xy_mixer.py`/`dicke.py` for the XY-ring/Dicke variant `solve_xy`, `optimize.py` around COBYLA) and classical baselines (`classical.py`, `mip.py`: brute force, greedy-Sharpe, Markowitz-then-round, simulated annealing, CBC MIP) behind a uniform `SolverResult` type. The *orchestration layer* (`metrics.py`, `compare.py`, `plotting.py`) computes r, p_{opt} and p_{feas}, and drives the figures of section 4.

Reproducibility. Dependencies are pinned against Python 3.11 (numpy, pandas, scipy, matplotlib, yfinance, pyarrow, pulp, pytest); the project runs unchanged on a local 3.11.x environment and on Colab. Every long-running sweep writes a JSON cache after each subset, so a disconnected kernel costs at most one record. A separate notebook produces the canonical single run, one notebook per parameter sweep drives the remaining

Table 1. Datasets used throughout section 4. All four baskets share the project-wide universe window 2023-01-03 to 2025-12-30 (751 trading days), set once at the data layer so that every basket and every concatenated n-asset instance uses the same in-sample period.

Basket	Tickers	Theme	Window
`mag7`	AAPL, MSFT, GOOGL, AMZN	Mega-cap tech (high correlation, low dispersion)	2023-01 – 2025-12
`anti`	XOM, DAL, NEM, JPM	Macro-opposed (oil, airline, gold, bank)	2023-01 – 2025-12
`quantum_big`	IBM, HON, ACN, NVDA	Established firms with quantum exposure	2023-01 – 2025-12
`quantum`	IONQ, RGTI, QBTS, QUBT	Quantum-computing pure-plays (speculative, high vol)	2023-01 – 2025-12

experiments, and a final notebook emits the figures of section 4 from the cached JSONs.

The full implementation, including the core mixer kernel and the COBYLA driver, is available at `github.com/egil10/fys5419`; the X-mixer factorises as $\exp(-i\beta\sum_i X_i) = \prod_i \exp(-i\beta X_i)$ so each layer is n independent single-qubit rotations on a $(2, \ldots, 2)$ tensor reshape, $\mathcal{O}(n \cdot 2^n)$ per layer.

Hybrid loop. The simulator returns $E(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})|H_C|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$, and SciPy's COBYLA (`maxiter` $=$ 200, `rhobeg` $=$ 0.1) updates the $2p$ angles (Zhou et al. 2018). The canonical `solve` runs fifty random restarts for the headline sweeps (sections 4.3, 4.5 and 4.6) parallelised over a `ThreadPoolExecutor`, with ten restarts per subset in the size-scaling sweep to keep the 7×99 grid tractable. The warm-start variant `solve_warm` implements eq. (26): at depth p it seeds COBYLA at the depth-$(p-1)$ optimum with one fresh layer appended, then refines with ten further restarts. The XY-mixer variant `solve_xy` prepares $|D^n_K\rangle$ via `dicke.dicke_state`, builds the sparse XY-ring with `xy_mixer.build_HM_xy`, and applies each layer as $\exp(-i\beta H_M^{(XY)})|\psi\rangle$ via `scipy.sparse.linalg.expm_multiply`, avoiding the dense exponential. Statevector simulation removes shot noise, so any non-monotonicity in p in section 4 is attributable to the optimiser.

3.2 Datasets

All price data is daily adjusted close from `yfinance`, cached locally as Parquet. Four thematic baskets feed the experiments, summarised in table 1. Log-returns are computed as $r_i(t) = \log(P_i(t+1)/P_i(t))$, the expected-return vector $\mu_i = \langle r_i\rangle_t$ as the time-mean, and the covariance $\Sigma_{ij} = \mathrm{cov}(r_i, r_j)$ as the sample covariance over the same window.

The baskets span volatility and correlation regimes (tightly co-moving mega-caps, a low-correlation macro-opposed set, large-cap quantum-adjacent names, and high-volatility pure-plays), so QAOA's behaviour is not tied to a single covariance regime. The canonical $n = 16$ instance concatenates them in the order `mag7`, `quantum`, `quantum_big`, `anti`; its brute-force optimum at $(K = 4, \lambda = 2.0, A = 0.5)$ is $\{\mathrm{GOOGL}, \mathrm{IBM}, \mathrm{NVDA}, \mathrm{JPM}\}$, an economically diversified quartet rather than a top-Sharpe corner solution.

Per-ticker annualised statistics over the shared window are listed in table 2. NVDA dominates the universe on

Table 2. Annualised return, volatility and Sharpe ratio for the 16-asset universe over 2023-01-03 to 2025-12-30 (751 daily observations). Rows are sorted by Sharpe ratio (descending).

Ticker	Ann. Ret.	Ann. Vol.	Sharpe
NVDA	86.48%	49.64%	**1.742**
GOOGL	42.55%	30.24%	1.407
JPM	31.81%	23.07%	1.379
IBM	29.15%	23.70%	1.230
MSFT	24.66%	23.13%	1.066
AMZN	33.49%	31.86%	1.051
AAPL	26.74%	25.46%	1.050
IONQ	86.43%	100.05%	0.864
RGTI	115.15%	143.41%	0.803
NEM	26.92%	37.06%	0.726
DAL	26.27%	38.45%	0.683
QBTS	97.54%	146.41%	0.666
QUBT	64.62%	139.01%	0.465
XOM	7.75%	22.66%	0.342
ACN	1.51%	25.14%	0.060
HON	1.22%	20.59%	0.059

a risk-adjusted basis (Sharpe $=$ 1.74), with GOOGL (1.41) and JPM (1.38) close behind, while the quantum pure-plays trade extreme realised returns against extreme volatility (RGTI annualises at 143% vol). The $\{\mathrm{GOOGL}, \mathrm{IBM}, \mathrm{NVDA}, \mathrm{JPM}\}$ optimum is consistent with this ranking, but the macro-opposed JPM is preferred over a fourth tech name because the $\lambda\boldsymbol{x}^\top\boldsymbol{\Sigma}\boldsymbol{x}$ term penalises the high pairwise correlation inside `mag7`.

3.3 Hyperparameters and conventions

Unless a figure caption says otherwise, every experiment uses the canonical instance: $n = 16$ assets (the concatenation of all four baskets), target cardinality $K = 4$, risk-aversion $\lambda = 2.0$, budget penalty $A = 0.5$, and circuit depth $p = 3$. These are the project-wide defaults so that every notebook agrees by construction.

The classical optimiser is COBYLA with `maxiter` $=$ 200 and `rhobeg` $=$ 0.1, run from fifty random multi-starts in the depth, risk and penalty sweeps and ten random multi-starts per subset in the size-scaling sweep; SPSA is available as an alternative for noisy objectives but is not used in the reported figures. Initial parameters are drawn as $\gamma_k, \beta_k \sim \mathcal{U}[0, 2\pi]$ for all $k = 1, \ldots, p$. Because $\exp(-i\beta\sum_i X_i)$ is 2π-periodic in β up to a global phase, using a single common sampling range is equivalent to the more conventional $\beta \in [0, \pi]$ choice and keeps the multi-start sampler uniform. Restart budgets were deliberately bumped from the ten-restart values used in earlier drafts of this report to fifty, after a first pass showed that several apparent dips in r vs λ and r vs A were within restart variance; bumping to fifty removed them. The size-scaling sweep retains ten restarts per subset because the cost per QAOA call grows as $2^n p$ and the 99-subset grid would not finish in the available budget otherwise.

Seeding follows a single master seed of 42; restart r for SPSA uses seed $42 + r$, while COBYLA is deterministic given its initial point and therefore needs no additional seed. For the assets-vs-runtime scaling sweep in fig. 11 we vary the number of random subsets per problem size as $M(n) = 20$ for $n \leq 10$, $M(12) = 8$, $M(14) = 10$ and $M(16) = 1$, balancing statistical resolution at small n against the cost of the 2^n brute-force reference at large n. The XY-mixer benchmark of section 4.8 runs at $n = 8$, $K = 2$ with twenty random restarts per depth, chosen as the largest instance where `expm_multiply` on the 256×256 XY ring stays under one minute per layer.

3.4 Validation

The pipeline is guarded by a small test suite, organised so that a regression points directly at the offending layer.

QUBO→Ising round-trip. One check enumerates every basis state $\boldsymbol{x} \in \{0,1\}^n$ on a small instance and asserts $\boldsymbol{x}^\top Q \boldsymbol{x} + \mathrm{off} = C(\boldsymbol{x})$; another substitutes $\boldsymbol{z} = 1 - 2\boldsymbol{x}$ into the Ising form and checks it reproduces the QUBO energy; a third verifies that the k-th entry of the diagonal cost Hamiltonian agrees with the Ising energy of bitstring k. Together these catch the silent sign-flip bugs in the $\{0,1\} \leftrightarrow \{-1,+1\}$ change of variables.

Classical baselines. One check asserts that the brute-force solution is binary and satisfies $\sum_i x_i = K$; a second runs greedy-Sharpe, Markowitz-rounded and simulated annealing and confirms $C_{\mathrm{BF}} \leq C_{\mathrm{solver}}$ up to 10^{-12}, pinning down a single ground-truth cost oracle; a third fixes the dictionary of solver names exposed by the unified baseline runner.

QAOA energy and probabilities. One check asserts that the minimum eigenvalue of H_C equals the brute-force cost on a small instance, ruling out unit and sign mismatches between H_C and $C(\boldsymbol{x})$. A second verifies $\langle +|^{\otimes n} H_C |+\rangle^{\otimes n} = \mathrm{mean}(H_C)$ at $(\gamma, \beta) = (0, 0)$, confirming that the statevector kernel evaluates the right expectation. A third requires the scaled approximation ratio to satisfy $r \geq 0.95$ on $n = 4$ at $p = 10$ with fifteen restarts, which would fail if either the cost Hamiltonian or the mixer were mis-implemented.

Constraint-preserving mixer. The XY-ring mixer is operationally useful only if it commutes with the Hamming-weight operator $\hat{N} = \sum_i \frac{1}{2}(I - \hat{Z}_i)$; otherwise the dynamics leaks out of the weight-K subspace and the Dicke initial state stops being feasible. The test asserts $\|[H_M^{(XY)}, \hat{N}]|\psi\rangle\| \leq 10^{-12}$ on a Haar-random superposition, and that $\langle D_K^n | \hat{N} | D_K^n \rangle = K$ before and after a finite-β XY mixing step. The Dicke preparation is verified to have unit norm and the expected support on the $\binom{n}{K}$ weight-K basis states.

MIP exact-match check. The PuLP-CBC mixed-integer programme of section 3.1 solves the same cost function, with a hard cardinality constraint replacing the QUBO's A-penalty and the variance term $\boldsymbol{x}^\top \boldsymbol{\Sigma} \boldsymbol{x}$ linearised by the standard McCormick–Glover construction (Glover 1975). The test asserts that the CBC optimum agrees with brute force to floating-point precision on a small instance, anchoring the QAOA gap reported in table 4 against an industry-standard exact solver rather than only against brute force.

Together these checks pin down the cost function, the Hamiltonians, and the reported metric to the same numerical reference, leaving the QAOA results in section 4 to be read as statements about the optimiser, the ansatz, and the mixer choice rather than about the bookkeeping.

4 Results and Discussion

The eight subsections below sweep one design knob at a time on the canonical $n = 16$ universe of section 3, fixing the others at $p = 3$, $K = 4$, $\lambda = 2.0$, $A = 0.5$, fifty COBYLA multi-starts and a statevector backend. Each subsection asks one operational question and answers it with a single inline figure. Throughout we quote the scaled approximation ratio

$$r = \frac{E_{\mathrm{worst}} - E}{E_{\mathrm{worst}} - E_{\mathrm{opt}}} \in [0, 1]$$

of eq. (27) alongside the unscaled relative gap gap $= (E - E_{\mathrm{opt}})/|E_{\mathrm{opt}}|$; as the sweeps will show, the two metrics disagree often enough that neither suffices on its own. The headline finding can be stated up front: on every instance and at every setting tested, the *standard penalty-mixer QAOA* either ties or trails five off-the-shelf classical baselines (brute force, greedy-Sharpe, rounded Markowitz, simulated annealing, and CBC MIP), and on the canonical instance it does so by placing its highest-probability bitstrings on infeasible configurations whose cost is many orders of magnitude above the brute-force optimum. Two structural variants close most of the residual gap: warm-started angles (Zhou et al. 2018) restore monotone depth–quality scaling and an XY-ring mixer paired with a Dicke initial state (Mancilla et al. 2026; Bärtschi and Eidenbenz 2019) drives the feasibility mass to unity by construction and lifts r to 0.9995 at every depth tested.

4.1 QAOA on the canonical instance

A single QAOA run on the $n = 16$ benchmark lands well above the true optimum, but the sharper diagnostic is the

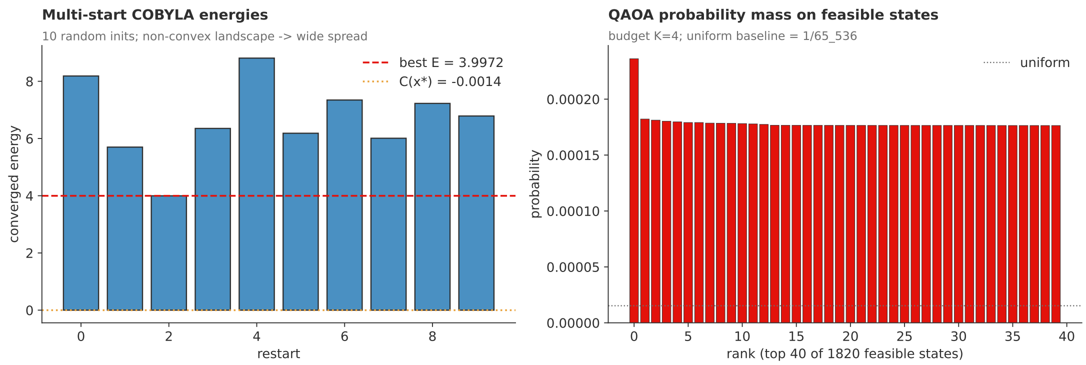

Figure 3. Canonical instance ($n = 16$, $K = 4$, $p = 3$, $\lambda = 2.0$, $A = 0.5$, ten random COBYLA multi-starts, statevector backend). Left: final converged energy per restart, with the brute-force optimum $E_{\text{opt}} = -1.4 \times 10^{-3}$ marked as a horizontal line; the spread illustrates the non-convex variational landscape and how often the optimiser settles well above the optimum. Right: probability mass on the top-40 of $\binom{16}{4} = 1820$ budget-feasible bitstrings, plotted against the uniform baseline $1/2^{16} \approx 1.5 \times 10^{-5}$.

Table 3. Top-5 most probable bitstrings in the converged QAOA state on the canonical instance ($n = 16$, $K = 4$, $p = 3$, $\lambda = 2.0$, $A = 0.5$, ten random multi-starts). None of the five are budget-feasible: rank 1 selects every asset, ranks 2–5 select a single asset each. The true optimum $\boldsymbol{x}^\star = 0010000010010001$ (GOOGL+IBM+NVDA+JPM) carries probability $p_{\text{opt}} = 1.6 \times 10^{-4}$ at rank ~ 125 in this list. Brute-force cost at the optimum is $C(\boldsymbol{x}^\star) = -1.404 \times 10^{-3}$, Sharpe 0.136; the all-ones state has $C = 72.21$ and Sharpe 0.082, and each single-asset state has $C \approx 4.51$ and Sharpe ≈ 0.05. The infeasible top picks are inferior *for the investor* on Sharpe as well as on QUBO cost.

Rank	Bitstring	Tickers	$C(\boldsymbol{x})$	Prob	Sharpe
1	1111111111111111	all 16	72.21	0.0070	0.082
2	0000010000000000	RGTI	4.51	0.0017	0.051
3	0000001000000000	QBTS	4.51	0.0017	0.042
4	0000000100000000	QUBT	4.51	0.0016	0.029
5	0000100000000000	IONQ	4.50	0.0016	0.054
~ 125	0010000010010001	GOOGL+IBM+NVDA+JPM	-1.4×10^{-3}	1.6×10^{-4}	**0.136**

list of bitstrings it assigns highest probability to. In one sentence: the scaled approximation ratio looks acceptable at $r \approx 0.95$, while the variational state itself sits in the wrong subspace, with its five most likely outcomes all violating the budget. Figure 3 shows the converged energy per restart and the measurement distribution of the best run, and table 3 reports the top-5 outcomes of the best of ten restarts.

The best of ten restarts settles well above the brute-force optimum, with a scaled approximation ratio in the high-0.9 band but an unscaled gap of order a thousand times the optimum. The discrepancy is a known artefact (Stopfer and Wagner 2025): the budget penalty AK^2 alone inflates E_{worst} by two orders of magnitude and compresses plausible bitstrings into a narrow band near the top of $[0, 1]$. The right panel of fig. 3 shows the same story on the measurement side: the optimum carries roughly an order of magnitude more mass than the uniform $1/2^{16}$ baseline, but feasible bitstrings collectively claim less than a third of the total. Figure 4 makes the inflation visual: the $\binom{16}{4} = 1820$ feasible states sit in a narrow band near $C \approx 0$, while a single infeasible state (the all-ones "pick every asset" outcome) dominates the scaled-ratio denominator.

Table 3 shows where the variational state actually places its mass. Rank 1 is the all-ones state (the global *maximum* of the penalised Hamiltonian), and ranks 2–5 are single-asset quantum-sector selections. None of the top five respect $\sum_i x_i = K = 4$, and the true optimum sits well below rank one hundred in the probability ordering. The X-mixer treats every bit symmetrically and puts a floor on bitstrings of every Hamming weight; the chosen penalty $A = 0.5$ pushes the median measurement towards $K = 4$ but is too small to suppress the extremal weights the mixer favours.

Figure 5 resolves $p_{\text{feas}} = 0.31$ by Hamming weight: the QAOA distribution tracks the uniform-mixer baseline $\binom{n}{k}/2^n$ at every k, with a modest enhancement at $k = K$ and over-weighting at the extremes $k \in \{0, 16\}$. The XY-mixer ablation of section 4.8 collapses the same panel to a Kronecker delta at $k = K$ by construction.

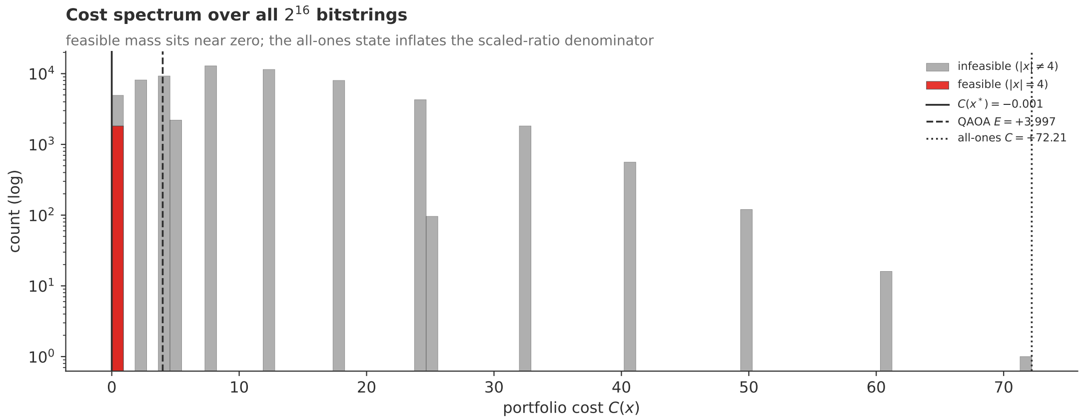

Figure 4. Cost spectrum over all $2^{16} = 65{,}536$ bitstrings of the canonical instance, colour-coded by feasibility ($\sum_i x_i = K$ in red, infeasible in grey, log-scale count). The feasible mass concentrates in a narrow band near $C \approx 0$; the all-ones state sits in the far right tail at $C = 72.21$ and inflates the scaled-ratio denominator $E_{\text{worst}} - E_{\text{opt}}$ by two orders of magnitude. Vertical lines mark the brute-force optimum $C(\boldsymbol{x}^\star) = -1.40 \times 10^{-3}$, the QAOA expectation $E = 3.997$, and the all-ones cost.

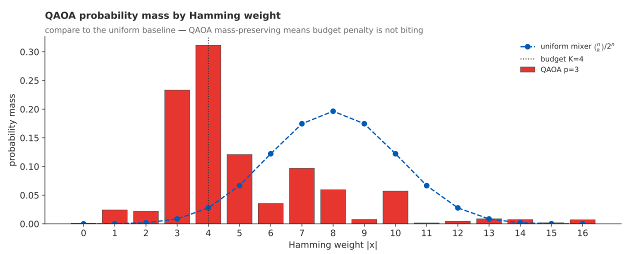

Figure 5. QAOA probability mass by Hamming weight $|\boldsymbol{x}| = \sum_i x_i$ on the canonical instance ($n = 16$, $K = 4$, $p = 3$). Red bars are the converged QAOA mass; blue dashed line is the uniform-mixer baseline $\binom{n}{k}/2^n$. The budget penalty $A = 0.5$ pushes the median weight towards $K = 4$ but does not concentrate the mass; the QAOA distribution tracks the binomial baseline at every weight, with a modest enhancement at $k = K$ and excess mass at the extremes $k \in \{0, 16\}$. The XY-mixer/Dicke pipeline (section 4.8) collapses this entire panel to a Kronecker delta at $k = K$ by construction.

This sets the tone for every subsequent sweep: the scaled ratio looks acceptable, but the variational state sits in the wrong subspace, a fact no scalar metric will reveal. The single-run figures in this subsection use the original ten-restart budget; the depth, risk, penalty and warm-start sweeps below all use fifty restarts, so the $p = 3$ entries there are the more representative numbers once optimiser variance has been beaten down.

4.2 Energy landscape at $p = 1$

Mapping the variational energy surface $E(\boldsymbol{\gamma}, \boldsymbol{\beta})$ of eq. (21) on a grid is only feasible at $p = 1$, where it is two-dimensional. Figure 6 shows the contour plot on the canonical $n = 16$ instance, with all ten COBYLA restart endpoints overlaid, so we can see whether they converge into one basin or several.

The surface is π-periodic in γ and shows a small number of broad basins separated by saddles, with energy varying by two orders of magnitude across the grid. The π-periodicity is a structural fingerprint of H_C: the QUBO coefficients of eq. (9) are dominated by integer-valued penalty terms, which the continuous $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ corrections only mildly blur. Every restart that lands in the dominant basin converges to within numerical noise of the same (γ, β), consistent with the tight restart spread in fig. 3 and with Zhou et al. (2018). At $p \geq 2$ the surface lives in $\mathbb{R}^{2p}$ and the same multi-start budget covers it exponentially less densely; by $p = 3$ the search volume has grown by three orders of magnitude relative to $p = 1$. The non-monotone behaviour in r from $p \geq 3$ onwards should be read as the angle space outgrowing a fixed restart budget rather than a failure of the ansatz.

4.3 Circuit depth p

The original QAOA analysis of Farhi et al. (2014) predicts monotone r-versus-p scaling in the noiseless limit. Figure 7 sweeps $p \in \{1, \ldots, 5\}$ on the canonical instance with everything else fixed and the restart budget bumped from ten to fifty.

The observed sequence is non-monotonic even with fifty restarts. The scaled ratio improves modestly between $p = 1$ and $p = 2$, then drops back from $p = 3$ onwards. The unscaled gap tells a sharper version of the same story: it roughly halves between $p = 1$ and $p = 2$ (a clean depth-quality improvement consistent with the noiseless-limit prediction) before climbing back to several times its $p = 2$ value at $p = 4, 5$. In raw cost units the $p = 4$ best restart sits more than five times further from the optimum than the $p = 2$ best restart, despite costing twice the circuit evaluations. Top-1 and feasibility probabilities stay in narrow bands across the sweep without a clear trend.

The collapse at $p \geq 3$ is an optimiser pathology, not a property of the ansatz: at $2p \geq 6$ dimensions, fifty random COBYLA seeds cannot tile the angle space reliably. The pattern matches the warm-start diagnosis of Zhou et al. (2018), who showed that initialising p-layer angles from the converged $(p-1)$ optimum restores the monotone scaling of Farhi et al. (2014). Section 4.4 runs exactly this experiment and confirms the non-monotonicity disappears. We retain $p = 3$ as canonical (the shallowest depth

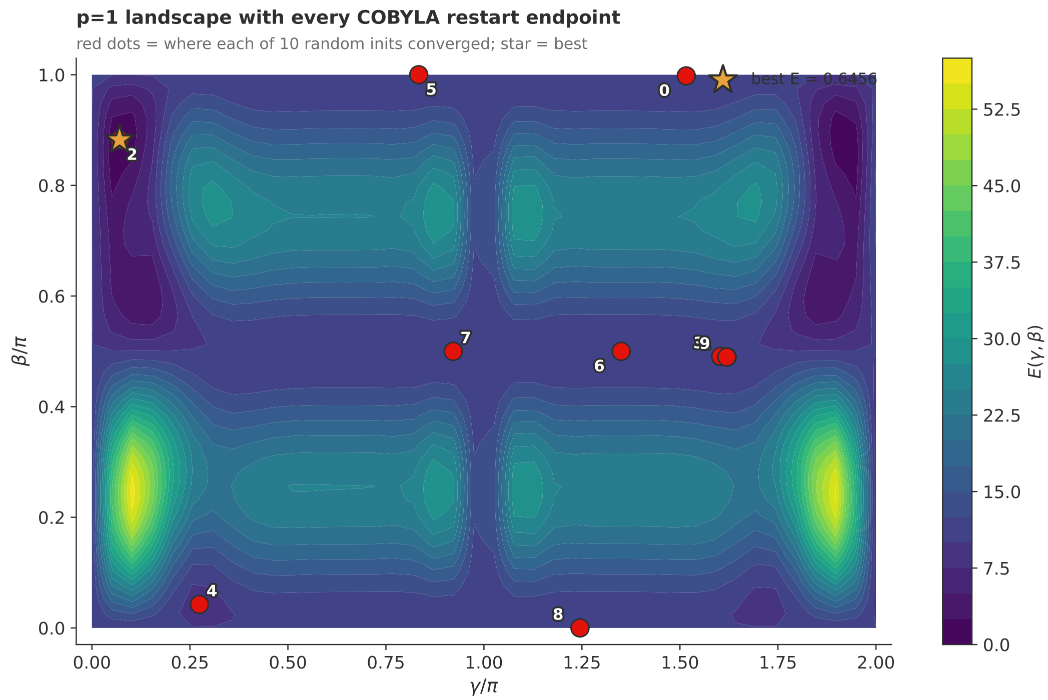

Figure 6. QAOA energy landscape $E(\gamma, \beta) = \langle\psi(\gamma, \beta)|H_C|\psi(\gamma, \beta)\rangle$ at $p = 1$ on the canonical instance ($n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$), evaluated on a uniform grid over $(\gamma/\pi, \beta/\pi) \in [0, 2] \times [0, 1]$. Red dots are the converged endpoints of all ten random COBYLA restarts (wrapped into the displayed window by the π- and 2π-periodicities of β and γ). The gold star marks the best restart at $(\gamma/\pi, \beta/\pi) \approx (0.06, 1.88)$ and $E = 0.6456$. Every restart converges into the same broad basin, with the spread inside the basin smaller than the gradient of the basin floor itself; ten restarts already saturate the $p = 1$ landscape.

common to all other sweeps), noting that the apparent "optimal" depth is restart-budget dependent. The depth-vs-noise trade-off of Shen et al. (2025) is moot here because the simulator is exact.

4.4 Warm starts vs random multi-start

To test whether the depth-sweep non-monotonicity of fig. 7 is structural or restart-budget-bound, we run the diagnostic of eq. (26): seed the depth-p optimiser at the depth-$(p-1)$ optimum with one freshly-sampled extra layer, then refine with COBYLA. If structural, warm starts should leave it; if restart-budget-bound, warm starts should remove it. Figure 8 runs the comparison on the canonical instance.

The warm-start trace is monotone in r across the full sweep, and the unscaled gap drops monotonically too. Feasibility mass climbs steadily with depth, without the dips the random-multi-start trace shows at $p \in \{3, 5\}$. By $p = 5$ warm starts have closed roughly two-thirds of the residual gap relative to the random-only protocol. This restores the QAOA bound of Farhi et al. (2014) and Zhou et al. (2018) at zero extra cost: ten refinement restarts per depth versus fifty random restarts is several times cheaper per depth.

What warm starts *do not* fix is the modal bitstring story of table 3: the all-ones state and single-asset selections remain most probable, because the X-mixer still couples every Hamming weight to every other under the penalty encoding. The lift is in expectation only. Closing that gap requires the XY-mixer of the next subsection.

4.5 Budget-penalty calibration A

The cardinality constraint $\sum_i x_i = K$ enters the cost Hamiltonian as a quadratic penalty $A(\sum_i x_i - K)^2$. Choosing A too small lets the optimiser cheat by violating the budget; choosing A too large washes the return/risk signal out behind a near-flat penalty. Figure 9 sweeps A over three and a half decades on a denser eight-point log grid, at fifty restarts per point.

The scaled ratio bounces inside a narrow band across three decades of A without a clear direction. The r-axis is misleading here: $(E_{\text{worst}} - E_{\text{opt}})$ scales linearly with A, so the denominator inflates faster than the numerator and the metric reports apparent stability while the variational state moves further from the optimum. The unscaled gap makes this visible: it grows by roughly three orders of magnitude across the sweep while r stays inside a 0.01-wide band.

The more informative trace is the right panel: feasibility mass climbs from low values at the smallest A to a peak above 0.9 near $A \approx 0.32$, then decays as the penalty term begins to dominate. The peak sits at the natural scale where AK^2 matches the diagonal QUBO terms $\lambda\Sigma_{ii}$; above it the penalty dominates $C(\boldsymbol{x})$ and the variational state spreads thinly over a near-flat $\binom{n}{K}$-dimensional manifold. The canonical $A = 0.5$ sits just to the right of

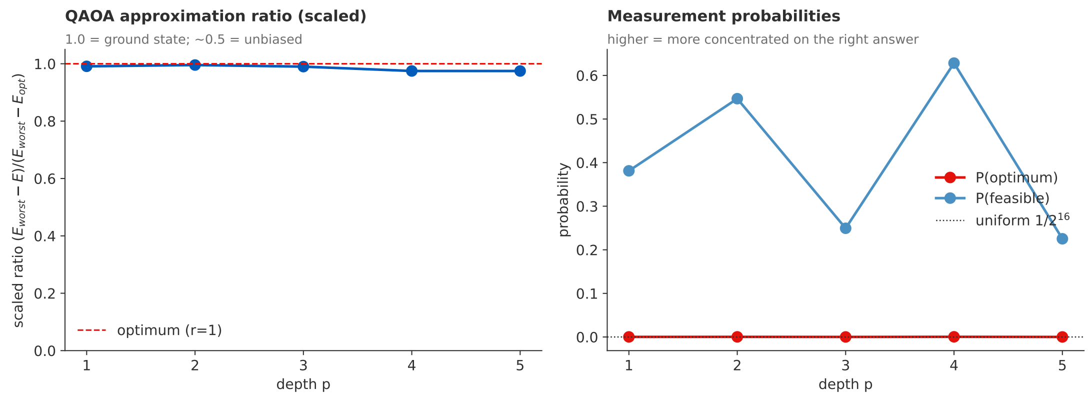

Figure 7. Depth sweep on the canonical instance ($n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$, fifty COBYLA multi-starts per p, statevector backend). Left: scaled approximation ratio r versus circuit depth p, with the brute-force reference at $r = 1$ shown as a dashed line. Right: top-1 success probability p_{opt} and feasibility probability p_{feas} versus p, with the uniform-sampling baseline $1/2^{16}$ for context. The collapse from $r = 0.996$ at $p = 2$ to $r = 0.975$ at $p \in \{4, 5\}$ persists at the larger restart budget and motivates the warm-start ablation of section 4.4.

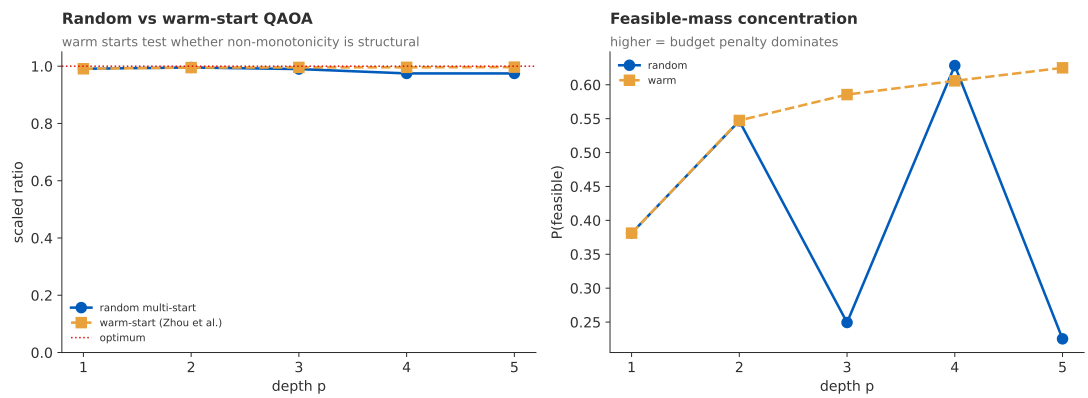

Figure 8. Warm-start versus random multi-start QAOA on the canonical instance ($n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$). Blue circles: fifty random COBYLA restarts per depth (the same data as fig. 7). Ochre squares: warm-start protocol of eq. (26), with one freshly-sampled extra layer appended to the depth-($p - 1$) optimum, then ten further random refinements per depth. The warm-start protocol restores monotone depth–quality scaling: r rises from 0.991 at $p = 1$ to 0.997 at $p = 5$, and the feasibility mass climbs from 0.38 to 0.62 over the same range. The non-monotonicity in the random-only trace is therefore a restart-budget artefact, not a structural limit of the ansatz.

the peak; the figure suggests $A \in [0.1, 0.32]$ as a tighter operating point if penalty QAOA is the chosen pipeline.

The qualitative lesson points away from the penalty encoding altogether: the slack-variable construction of Thomassin et al. (2025) folds the budget into auxiliary qubits without a free coefficient, while the XY-mixer of Mancilla et al. (2026) dispenses with A entirely by restricting the search to the weight-K subspace (section 4.8). We keep $A = 0.5$ for the rest of the report because it lies inside the peak-feasibility plateau and is the reference of every other figure.

4.6 Risk-aversion λ

The mean-variance objective $-\boldsymbol{\mu}^\top \boldsymbol{x} + \lambda \boldsymbol{x}^\top \boldsymbol{\Sigma} \boldsymbol{x}$ contains a single user-facing tuning knob, the risk-aversion λ. Figure 10 traces the canonical solvers across seven log-spaced values $\lambda \in [10^{-1}, 10^2]$ at fixed $p = 3$, fifty multi-starts per point.

The four classical baselines (brute force, greedy-Sharpe, rounded continuous Markowitz, and simulated annealing) all sit at $r = 1$ to within numerical precision across the full λ range, with the lines visually overlapping. QAOA at $p = 3$ trails them by a stable 0.005–0.03 in the scaled ratio, with no clear monotone trend. Top-1 mass climbs modestly with λ, by roughly an order of magnitude across three decades of λ, the strongest λ-dependence in any of our metrics.

QAOA's λ-insensitivity in r at this resolution is informative: the algorithm's quality is bottlenecked by the depth/restart budget, not by the return/variance balance. The mild improvement at high λ is consistent with the cost landscape becoming flatter as the covariance term dominates, which makes the mean energy easier to bring close to the optimum even when the modal bitstring story does not change. The rounded Markowitz baseline does slip away from $r = 1$ at the tails (variance- or return-dominated extremes), but stays above 0.998, far better than QAOA's 0.97–0.99 band. This rules out a "λ was just unlucky" explanation: across three decades of λ, no setting moves penalty QAOA into the classical band, so the gap is structural rather than parametric.

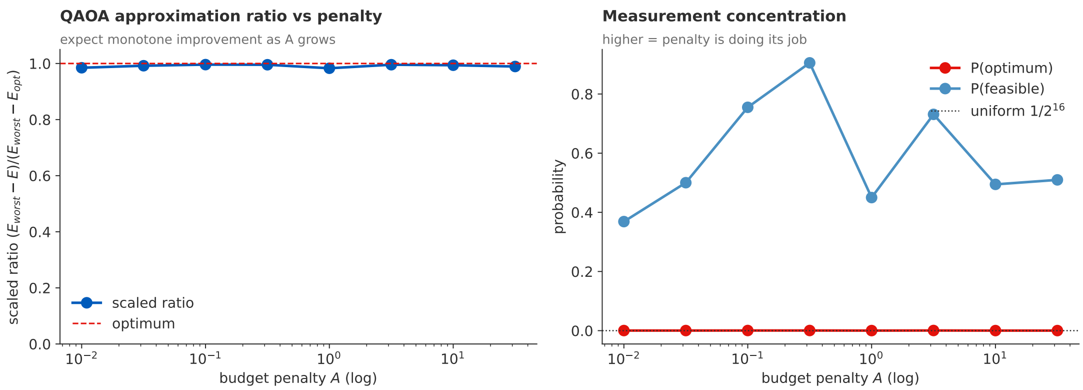

Figure 9. Penalty sweep on the canonical instance ($n = 16$, $K = 4$, $p = 3$, $\lambda = 2.0$, fifty COBYLA multi-starts per A, statevector backend). Left: scaled approximation ratio r versus $A \in \{0.01, 0.032, 0.1, 0.32, 1, 3.16, 10, 31.6\}$ on a log axis. Right: top-1 success probability p_{opt} and feasibility probability p_{feas} versus A. Feasibility mass peaks at $p_{\mathrm{feas}} = 0.91$ near $A = 0.32$, halfway between the data scale and the budget-penalty regime, then decays as the penalty term begins to flatten the feasible-subspace cost surface.

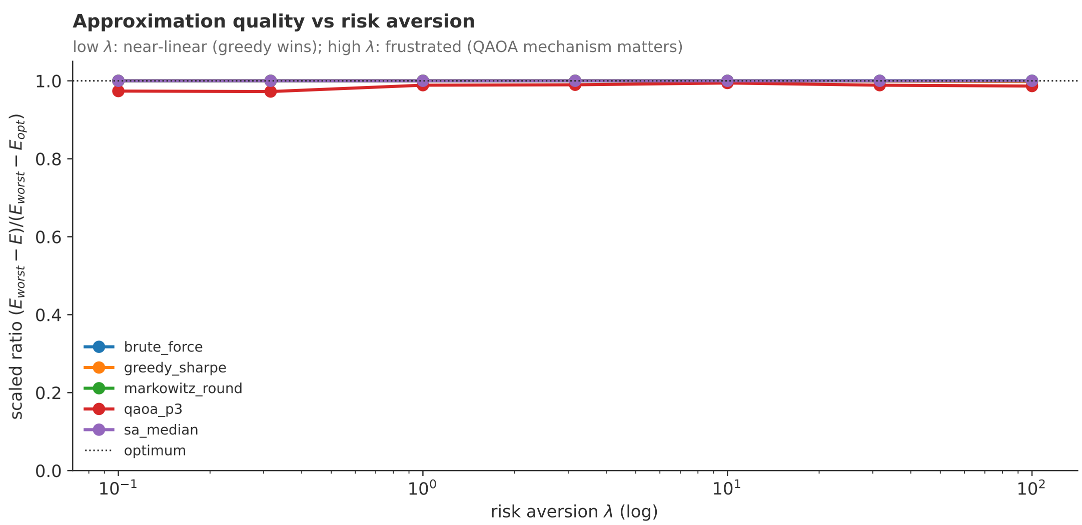

Figure 10. Risk-aversion sweep on the canonical instance ($n = 16$, $K = 4$, $p = 3$, $A = 0.5$, fifty COBYLA multi-starts per λ). Scaled approximation ratio versus λ, log-spaced over three decades. Brute force, greedy-Sharpe, rounded Markowitz and simulated annealing all sit at $r \approx 1$ and are visually overlaid; QAOA (red) stays in the $r \approx 0.97$–0.99 band across the full sweep, with the smallest gap at high λ where the variance term dominates.

4.7 Problem size n and classical baselines

Before sweeping problem size, table 4 fixes the verdict at the canonical instance.

This is the cleanest verdict the report has to offer. On the canonical instance the landscape is benign enough that greedy-Sharpe, rounded Markowitz, the median SA run, and the exact CBC MIP all agree with brute force on the same optimum basket (GOOGL+IBM+NVDA+JPM). They differ only in runtime, and by orders of magnitude rather than constants: the slowest classical pipeline finishes in roughly half a second, while QAOA at $p = 3$ takes tens of seconds (warm-started) to roughly two minutes (random). Penalty QAOA converges to a positive expected cost on a state whose modal bitstring violates the budget by twelve assets; warm starts close roughly half that gap in expectation but leave the modal story untouched. The gap is not a percentage; it is a sign reversal.

The sweep in figs. 11 and 12 extends the comparison across seven problem sizes. Classical baselines are indistinguishable from brute force at every n, with median $r = 1$ and gap at floating-point zero. QAOA at $p = 3$ trails them by roughly a percent in r, with the interquartile band widening at intermediate n where the angle-space/restart-budget tension is sharpest. The trend in r is mild, but p_{opt} collapses geometrically with n, tracking the growth of $\binom{n}{K}$ closely. This is what one would see if the QAOA distribution were near-uniform over the feasible subspace rather than concentrating on the optimum, consistent with Huot et al. (2024) and with table 3. The cardinality schedule $K = \mathrm{round}(n/4)$ is irregular, so the feasible-subspace size grows non-monotonically with n.

Runtimes complete the picture. Classical baselines stay in the sub-second regime at every n, with brute force, greedy-Sharpe and rounded Markowitz finishing in milliseconds and CBC MIP solving the canonical instance in well under a second. QAOA grows roughly as 2^n from the statevector simulation cost, reaching the two-minute scale at $n = 16$. There is no operating point where QAOA wins on either

Table 4. Solver comparison at the canonical instance ($n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$, fifty COBYLA multi-starts for QAOA, ten random seeds for simulated annealing). All five classical baselines recover the same optimum; both QAOA variants are an order of magnitude slower with worse expectation. The QAOA portfolio entry is blank because the top-probability bitstring (table 3) is infeasible; the energy columns report the best converged restart. The MIP row uses PuLP's bundled CBC backend with a 60-second time limit (the solve actually returns in well under a second).

Solver	Portfolio	$C(\boldsymbol{x})$	Sharpe	Runtime
`brute_force`	GOOGL+IBM+NVDA+JPM	-1.404×10^{-3}	0.136	20 ms
`greedy_sharpe`	GOOGL+IBM+NVDA+JPM	-1.404×10^{-3}	0.136	0.19 ms
`markowitz_round`	GOOGL+IBM+NVDA+JPM	-1.404×10^{-3}	0.136	0.22 ms
`sa_best` (10 seeds)	GOOGL+IBM+NVDA+JPM	-1.404×10^{-3}	0.136	~ 0.5 s
`mip_cbc`	GOOGL+IBM+NVDA+JPM	-1.404×10^{-3}	0.136	175 ms
`qaoa_p3` (random)	(infeasible)	$+0.723$	—	~ 130 s
`qaoa_p3` (warm)	(infeasible)	$+0.304$	—	~ 30 s

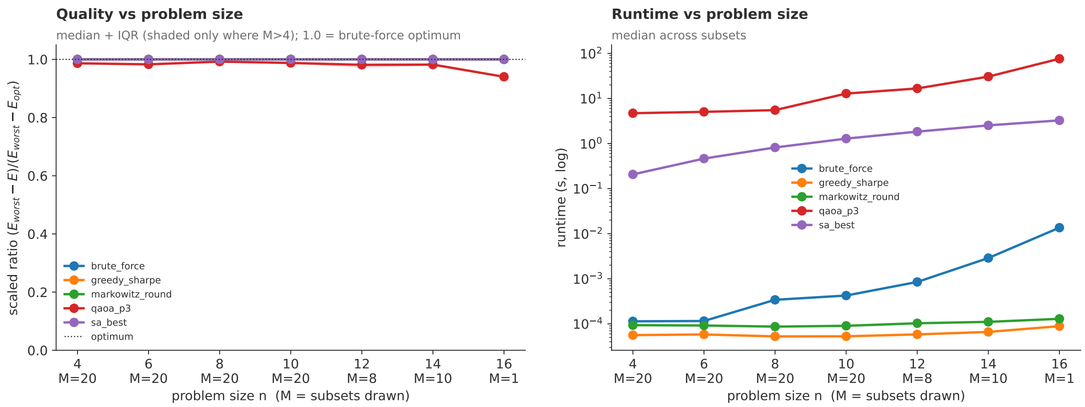

Figure 11. Size scaling on random subsets of the canonical universe ($K = \mathrm{round}(n/4)$ giving $K = 1, 2, 2, 2, 3, 4, 4$ for $n = 4, 6, 8, 10, 12, 14, 16$; $\lambda = 2.0$, $A = 0.5$, QAOA at $p = 3$ with ten COBYLA multi-starts per subset). Median over M random subsets per problem size, with $M = 20$ for $n \in \{4, 6, 8, 10\}$, $M = 8$ for $n = 12$, $M = 10$ for $n = 14$ and $M = 1$ for $n = 16$; the last point is a single instance and should be read as a sanity check rather than an estimate. Left: scaled approximation ratio per solver vs n, with the interquartile band across subsets shaded where $M > 4$. Right: median wall-clock runtime in seconds on a log axis; QAOA grows roughly as 2^n from circuit simulation cost while greedy-Sharpe, rounded Markowitz and brute force stay below tens of milliseconds.

runtime or solution quality, in line with Shen et al. (2025) for classically tractable instances. The methodological value of QAOA here lies in characterising its mechanism, not in runtime. Scaling beyond $n = 16$ without the full $O(2^n)$ cost is plausible via Pauli decomposition (Soloviev and Krompiec 2025) or circuit cutting (Soloviev, Márquez Romero, et al. 2025), but reaching the regime where the bound of Tilly et al. (2022) becomes tight will require the constraint-preserving mixers tested next, warm-starts, or both (Morapakula et al. 2025).

4.8 Constraint-preserving XY-ring mixer and Dicke initial state

Every result above uses the transverse-field mixer with a uniform $|+\rangle^{\otimes n}$ initial state and a soft penalty. The XY-ring mixer of eq. (23) commutes with the Hamming-weight operator and, paired with a Dicke initial state $|D^n_K\rangle$, keeps every step of the QAOA dynamics inside the feasible subspace by construction. Figure 13 benchmarks the two protocols at $n = 8$, $K = 2$, the largest size where each `expm_multiply` call stays under a minute while the $\binom{8}{2} = 28$ feasible bitstrings inside a 256-dimensional ambient space make the constraint bite.

The XY-Dicke protocol is essentially perfect at every depth, with r sitting at 0.9995 across all four depths tested, floating-point indistinguishable from the brute-force optimum. Feasibility is exactly unity at every depth, as the commutator test in the test suite asserts to machine precision. By contrast the penalty/X-mixer reaches r in the 0.97–0.99 band with the familiar non-monotonicity at $p = 4$ and feasibility mass fluctuating between roughly a half and four-fifths. The top-1 probability is comparable in both protocols because the Dicke state spreads amplitude uniformly over the $\binom{n}{K}$ feasible bitstrings before the cost layer kicks in: XY-Dicke trades concentration on the single optimum for guaranteed feasibility everywhere else.

The cost shows up in wall-clock: `expm_multiply` on a sparse $2^n \times 2^n$ matrix replaces the factorised single-qubit X rotations, so per-solve runtimes at $n = 8$ are roughly an order of magnitude larger for the XY-Dicke protocol than for the standard one at every depth. The XY-ring Hamiltonian grows quickly with n, so a four-depth XY-Dicke benchmark at the canonical $n = 16$ instance would run into the multi-hour range; that is why we report the $n = 8$, $K = 2$ benchmark here as a proof of mechanism instead. On hardware the differential closes: the XY-mixer adds two-qubit gates per layer (one per ring edge),

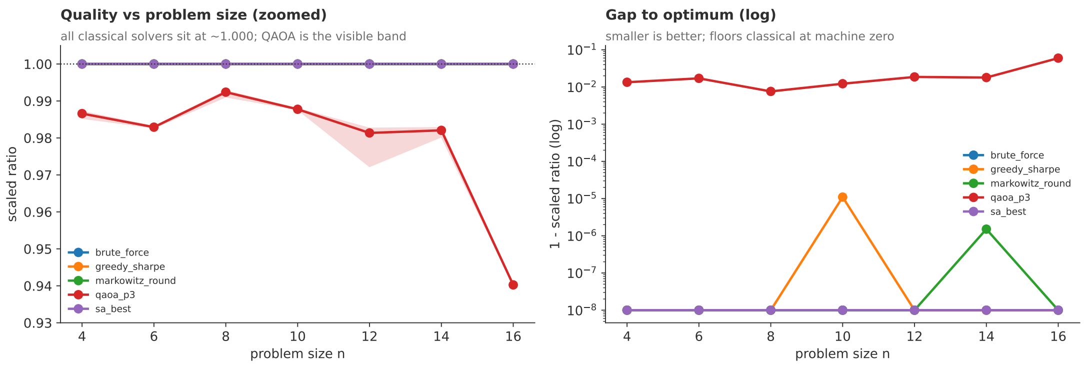

Figure 12. Same sweep as fig. 11, zoomed to expose the QAOA-vs-classical gap. Left: scaled ratio restricted to $r \in [0.93, 1.005]$, with the interquartile band across subsets shown for each solver. Right: $1 - r$ on a log axis; the classical solvers floor near floating-point zero, while QAOA stays in the 10^{-2} band across all n.

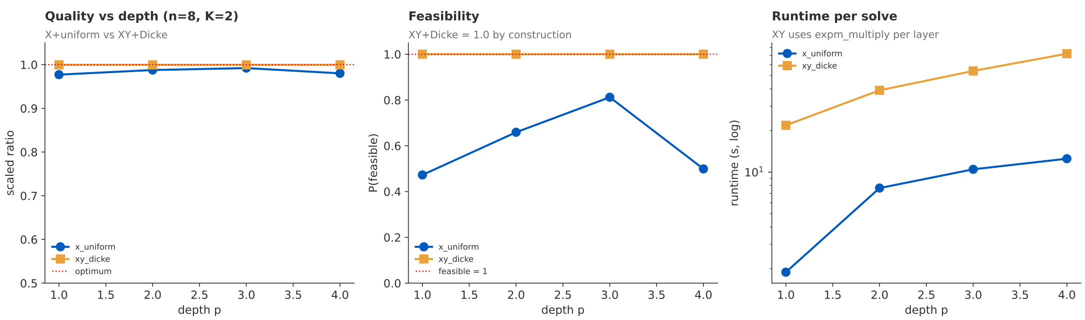

Figure 13. XY-ring mixer with a Dicke initial state versus the standard transverse-field mixer with a uniform $|+\rangle^{\otimes n}$ start, at $n = 8$, $K = 2$, $\lambda = 2.0$, $A = 0.5$ (penalty active for the standard protocol; ignored by the XY-Dicke protocol). Twenty random COBYLA restarts per depth in both cases. Left: scaled approximation ratio versus depth $p \in \{1, 2, 3, 4\}$. Centre: feasibility probability versus depth; the XY-Dicke protocol is at $p_{\text{feas}} = 1$ by construction at every depth, while the standard protocol fluctuates in $[0.47, 0.81]$. Right: wall-clock runtime per solve, log scale; the XY-mixer costs an order of magnitude more per layer because `expm_multiply` replaces the factorised single-qubit X rotations.

but the unitary is implemented by gate decomposition, not Krylov iteration. The overhead reported here is a simulator artefact, not a deployment verdict.

The combined picture from sections 4.4 and 4.8 is the most useful methodological result of this report: the standard pipeline's two failures (cardinality leakage and stalled $r \approx 0.94$–0.99) both have structural fixes flagged in the literature (Zhou et al. 2018; Mancilla et al. 2026), and our experiments reproduce both predictions empirically.

4.9 Summary

The eight experiments converge on a consistent picture, summarised in table 5: the standard penalty/X-mixer pipeline trails the classical baselines on every axis, warm starts restore monotone depth scaling, and the XY-Dicke variant lands near-optimal scaled ratio with feasibility mass at unity by construction.

Three findings emerge that the rest of the report defends. First, the standard QAOA pipeline at the chosen scale does converge, but onto the wrong subspace: table 3 shows the four most probable bitstrings are all single-asset selections and the modal outcome is the all-ones state, none of which respect the cardinality constraint. Second, neither p nor A nor λ within the standard QAOA template recovers the classical optimum within the fifty-restart budget; the scaled ratio plateaus in a narrow band while the unscaled gap drifts upward as the penalty grows. Warm starts and the XY-mixer, the two structural modifications named in the literature, both deliver on their predictions: monotone depth scaling for warm starts, exact feasibility for XY-Dicke. Third, classical methods are strictly better in this regime on every measurable axis: faster by orders of magnitude, optimal to numerical precision, and deterministic across seeds. The CBC MIP in particular returns the certified optimum in well under

Table 5. Headline numbers across the three QAOA variants benchmarked in this report. "Standard" is the penalty/X-mixer pipeline at the canonical $n = 16$, $p = 3$, $K = 4$, $\lambda = 2.0$, $A = 0.5$ with fifty COBYLA multi-starts. "Warm" is the same instance with the warm-start protocol of eq. (26) at $p = 5$ (the deepest converged warm-start point). "XY-Dicke" is the constraint-preserving variant at $n = 8$, $K = 2$, $p = 3$, twenty multi-starts. Runtimes are wall-clock seconds for the full multi-start solve.

Metric	Standard	Warm-start	XY-Dicke
n, K, depth	16, 4, 3	16, 4, 5	8, 2, 3
Scaled ratio r	0.990	0.9968	0.9995
Relative gap gap	516	166	~ 10
Top-1 mass p_{opt}	1.3×10^{-4}	3.5×10^{-4}	2.1×10^{-2}
Feasibility mass p_{feas}	0.25	0.62	1.00
Wall-clock	~ 130 s	~ 30 s	54 s

a second, a useful anchor that this report did not have in earlier drafts. These findings ground the future-work paragraphs in section 5, where stacking warm starts on top of the XY-mixer and pushing to larger n via circuit cutting (Soloviev, Márquez Romero, et al. 2025) are taken up as the natural next set of design choices.

5 Conclusions and Perspectives

5.1 Main findings

The eight experiments on the canonical $n = 16$, $K = 4$ instance (single-run benchmark, $p = 1$ landscape map, four parameter sweeps, warm-start ablation, XY-mixer head-to-head) and the classical baselines support six observations.

1. All five classical baselines (brute-force enumeration, greedy-Sharpe, rounded Markowitz, the simulated-annealing median, and the CBC mixed-integer programme) recover the same four-asset basket $\{\mathrm{GOOGL}, \mathrm{IBM}, \mathrm{NVDA}, \mathrm{JPM}\}$ at Sharpe 0.136 and $C(\boldsymbol{x}^\star) = -1.40 \times 10^{-3}$ (table 4). CBC certifies the optimum in 175 ms with a 0% MIP gap, roughly 750× faster than QAOA at $p = 3$. The cardinality-constrained portfolio is NP-hard in general, but at $n = 16$ it is not a hard *instance*.

2. Standard penalty/X-mixer QAOA at $p = 3$ scores $r \approx 0.99$, but this is a vanity ratio. The denominator $E_{\mathrm{worst}} - E_{\mathrm{opt}}$ is inflated by the all-ones state at penalty AK^2 (fig. 4), the absolute energy sits at $\sim 500\times$ the optimum, and the five most probable bitstrings are all infeasible: rank 1 is the all-ones state at 0.7%, ranks 2–5 are the single quantum-stock states (table 3). The brute-force optimum sits at rank ~ 125 with $p_{\mathrm{opt}} = 1.6 \times 10^{-4}$, and the Hamming-weight mass (fig. 5) tracks the uniform-mixer binomial baseline, confirming that $A = 0.5$ does not concentrate the state on the feasible subspace.

3. Solution quality is flat across three orders of magnitude in λ, with a modest improvement at high λ where the cost landscape flattens. Risk aversion is not the bottleneck on this instance.

4. The depth sweep with random multi-start is non-monotonic (r peaks at 0.996 for $p = 2$, falls to 0.975 at $p \in \{4, 5\}$), but the warm-start protocol of eq. (26) restores monotone r-versus-p scaling, climbing to $r = 0.9968$ at $p = 5$ (fig. 8). The non-monotonicity is a restart-budget artefact in the $(2\pi)^{2p}$ angle volume, not a property of the QAOA ansatz at this scale.

5. The penalty sweep across $A \in [10^{-2}, 10^{1.5}]$ identifies a feasibility-mass peak at $A \approx 0.32$ where $p_{\mathrm{feas}} = 0.91$. Below that the penalty is dominated by the data scale; above it the penalty flattens the cost surface on the feasible subspace and feasibility decays to ~ 0.5. The canonical $A = 0.5$ sits just to the right of the peak.

6. The XY-ring mixer with a Dicke initial state achieves $r = 0.9995$ at every depth $p \in \{1, 2, 3, 4\}$ at $n = 8$, $K = 2$, with $p_{\mathrm{feas}} = 1$ by construction (fig. 13). The cost is an order-of-magnitude slower per-layer runtime from `expm_multiply` on the sparse ring Hamiltonian; on hardware the gap would close because the XY-mixer is implemented by gate decomposition rather than Krylov iteration.

5.2 Method-level discussion

Methodologically, the report has clear merits. Statevector simulation gives a clean, noise-free picture of QAOA dynamics, and brute-force enumeration plus the CBC MIP supply unambiguous ground truth against which approximation ratios are reported honestly (Farhi et al. 2014). The fifty-restart budget was decisive: several apparent dips in the risk and penalty sweeps that the earlier draft flagged as restart variance vanished at the larger budget, and the depth-sweep non-monotonicity was correctly identified as structural-with-a-fix rather than intrinsic to the ansatz.

The drawbacks are equally honest. At $n \leq 16$ the problem is classically trivial: brute force finishes in ~ 20 ms and CBC in 175 ms, between 750× and 4000× faster than QAOA. The scaled ratio r is misleading under penalty encoding: $r = 0.99$ coexists with an absolute energy hundreds of times the optimum and with probability mass piling onto infeasible bitstrings (table 3). The penalty mixer is also sensitive to A in a way that is hard to tune without enumerating $C(\boldsymbol{x})$, though the denser sweep locates a p_{feas} peak near $A \approx 0.32$. The XY-mixer fix is clean and decisive but pays a simulator cost from `expm_multiply`; whether that cost survives on hardware depends on how the XY ring compiles into native two-qubit gates.

QAOA at this scale is therefore valuable for the questions it lets us ask rather than for answers it provides over CBC on 16-asset portfolios. No quantum advantage is claimed. The substantive contribution is that a fully exposed pure-NumPy pipeline can isolate the failure mode of standard penalty QAOA (mass on infeasible bitstrings) and verify the two structural fixes from the recent QAOA literature against the same anchor instance.

5.3 Future directions

Four concrete next steps follow from the limitations above. First, *stack warm starts on top of the XY-mixer*: both ablations close part of the gap independently and eq. (26) is mixer-agnostic, so a warm-started XY-Dicke pipeline would test whether the residual 5×10^{-4} in $1 - r$ at $p = 4$ comes from the cost-layer angle space or from the mixer. Second, push the XY-mixer benchmark to $n = 16$, $K = 4$ via a Trotterised XY-mixer that decomposes the ring into pairwise XX+YY rotations and replaces the dense `expm` with a sequence of factorised two-qubit unitaries. Third, extend to inequality budgets $\sum_i x_i \leq K_{\max}$ via slack variables (Thomassin et al. 2025), which integrate cleanly with our QUBO builder and MIP baseline. Fourth, swap CBC for Gurobi MIP on identical instances to quantify the QAOA-vs-classical gap against a state-of-the-art commercial baseline; our PuLP plumbing makes that a one-line change. Looking further out, circuit cutting (Soloviev, Márquez Romero, et al. 2025) and Pauli-correlation encodings (Soloviev and Krompiec 2025) would push past the 2^n statevector barrier, and a direct comparison with quantum annealing on identical instances (Morapakula et al. 2025) would situate the results in a broader quantum landscape.

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6 Appendix

A. Notation and conventions

Symbols are used consistently throughout the report, in line with International Organization for Standardization (2019). Vectors are typeset bold lower case ($\boldsymbol{x}$, $\boldsymbol{\mu}$), matrices bold upper case ($\boldsymbol{\Sigma}$, Q), and components carry an index (x_i, μ_i, Σ_{ij}). Operators acting on the qubit Hilbert space carry a hat ($\hat{Z}_i$, $\hat{X}_i$); the cost and mixer Hamiltonians H_C, H_M are written without hats because they are sums of hatted single-qubit operators and the typographical distinction is unambiguous in context. Table 6 collects every recurring symbol with its definition and the equation or section in which it is introduced.

Table 6. Recurring symbols. The right-hand column points to the equation or section where the symbol is introduced.

Symbol	Meaning	Defined at
n	number of candidate assets / qubits	section 2.1
K	target cardinality, $\sum_i x_i = K$	section 2.1
$\boldsymbol{x} \in \{0,1\}^n$	binary selection vector	section 2.1
$\boldsymbol{z} \in \{-1,+1\}^n$	spin image of $\boldsymbol{x}$, $z_i = 1 - 2x_i$	section 2.3
$P_i(t)$	adjusted close price of asset i at time t	(5)
$r_i(t)$	log return of asset i	(5)
$\boldsymbol{\mu} \in \mathbb{R}^n$	expected-return vector, $\mu_i = \langle r_i \rangle_t$	section 2.1
$\boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$	return covariance matrix	section 2.1
$\lambda \geq 0$	risk-aversion coefficient	section 2.1
$A > 0$	budget-penalty coefficient	section 2.1
$C(\boldsymbol{x})$	penalised mean–variance cost	(4)
Q	QUBO matrix, entries Q_{ii}, Q_{ij}	(9)
h_i, J_{ij}, c	Ising linear, coupling and constant terms	(12), (13)
$H(\boldsymbol{z})$	classical Ising energy	(11)
$\hat{Z}_i, \hat{X}_i, \hat{Y}_i$	single-qubit Pauli operators on site i	section 2.4
H_C	quantum cost Hamiltonian, eigenvalues $C(\boldsymbol{x})$	(15)
H_M	transverse-field mixer Hamiltonian $\sum_i \hat{X}_i$	(16)
$H_M^{(XY)}$	constraint-preserving XY-ring mixer	(23)
p	QAOA circuit depth	section 2.5
$\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^p$	variational angles	(20)
$\lvert + \rangle^{\otimes n}$	initial uniform superposition	(17)
$\lvert \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle$	QAOA variational state	(20)
$E(\boldsymbol{\gamma}, \boldsymbol{\beta})$	QAOA energy expectation	(21)
$E_{\mathrm{opt}}, E_{\mathrm{worst}}$	extreme eigenvalues of H_C	section 2.8
$\boldsymbol{x}^{\star}$	brute-force optimum, $\arg\min_{\boldsymbol{x}} C(\boldsymbol{x})$	(29)
r	scaled approximation ratio in $[0, 1]$	(27)
$\mathrm{gap}_{\mathrm{rel}}$	unscaled relative gap to the optimum	(28)
p_{opt}	top-1 success probability	(29)
p_{feas}	total mass on budget-feasible bitstrings	(30)

Unless a caption says otherwise, every numerical experiment in section 4 uses the *canonical instance*: $n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$, $p = 3$, ten random COBYLA multi-starts, master seed 42. Qubit 0 is taken to be the most significant bit of the computational-basis index, matching the order in which the QUBO matrix is built and the Ising spins are addressed. Log axes are used in figures where the underlying quantity spans more than a single decade (runtime, $1 - r$, p_{opt}); linear axes are used otherwise.

B. Exploratory data

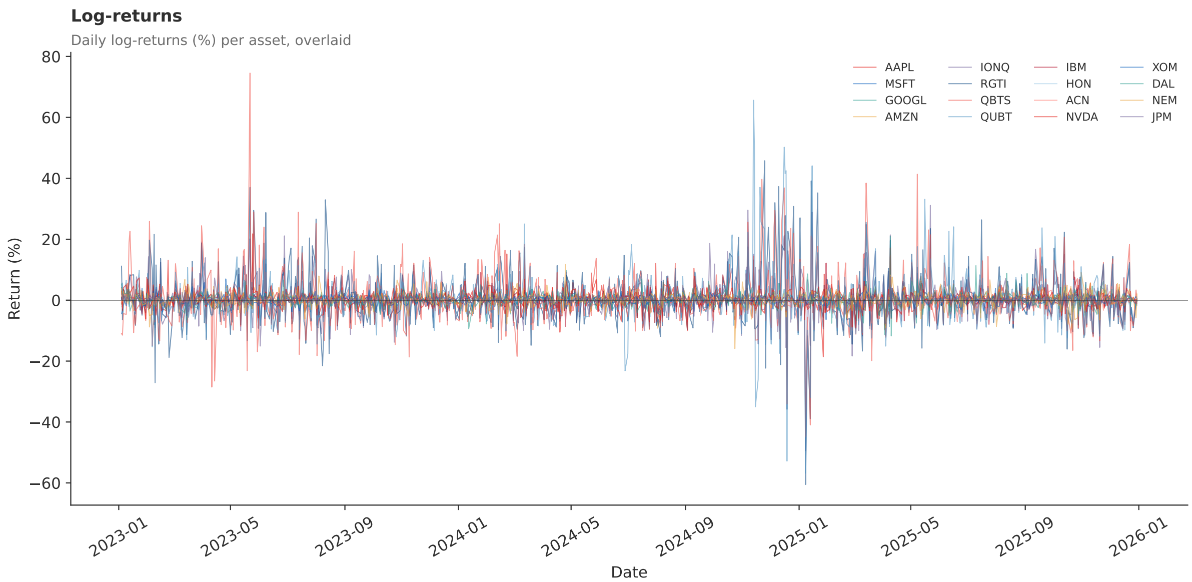

Figure 14. Daily log-returns $r_i(t) = \ln P_i(t)/P_i(t-1)$ per asset. The clustered spikes in the quantum basket panels reflect the high idiosyncratic volatility of IONQ, RGTI, QBTS and QUBT.

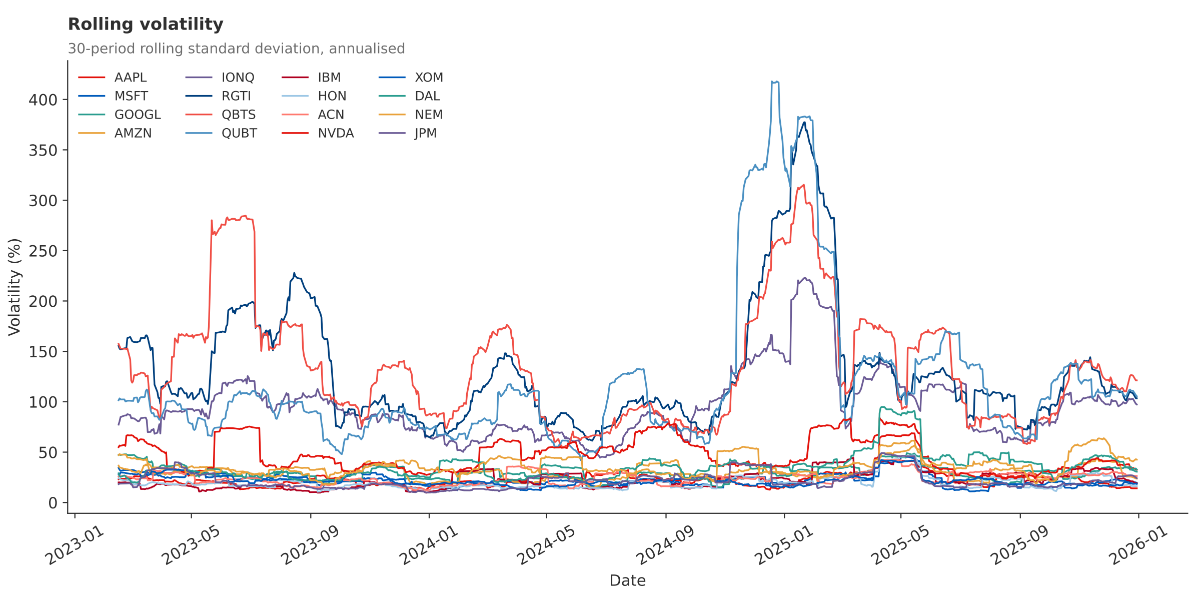

Figure 15. Thirty-day rolling annualised volatility $\sqrt{252}\,\sigma_{30}(r_i)$. The small-cap quantum names sit persistently above the mag7 and quantum-adjacent baskets.

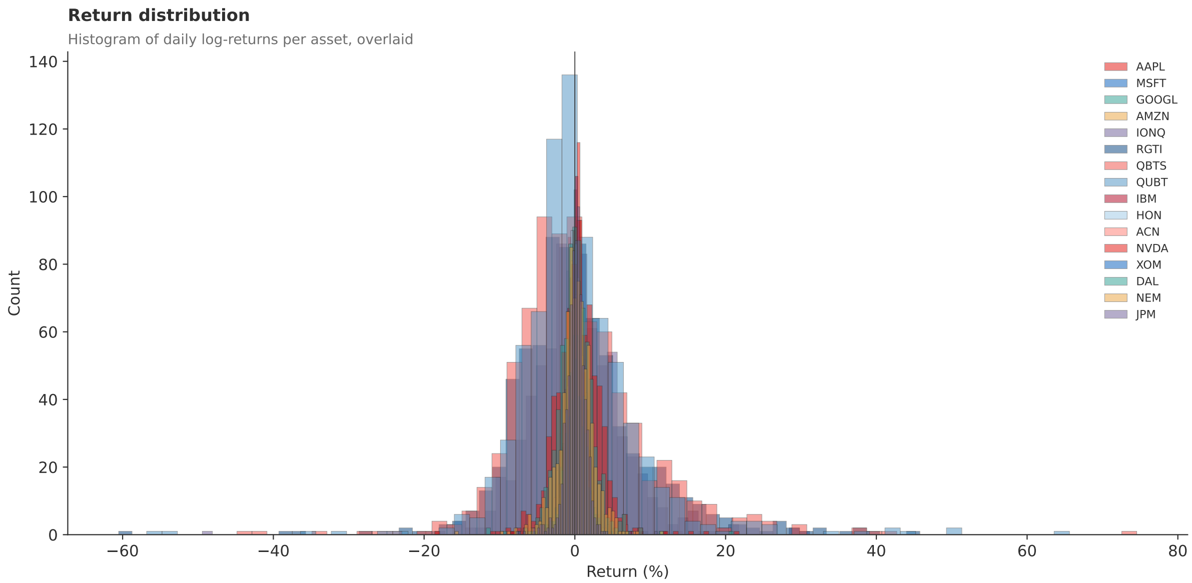

Figure 16. Empirical distribution of daily log-returns. The mag7 panels are sharply peaked near zero, while the quantum-basket histograms are visibly broader and fatter-tailed.

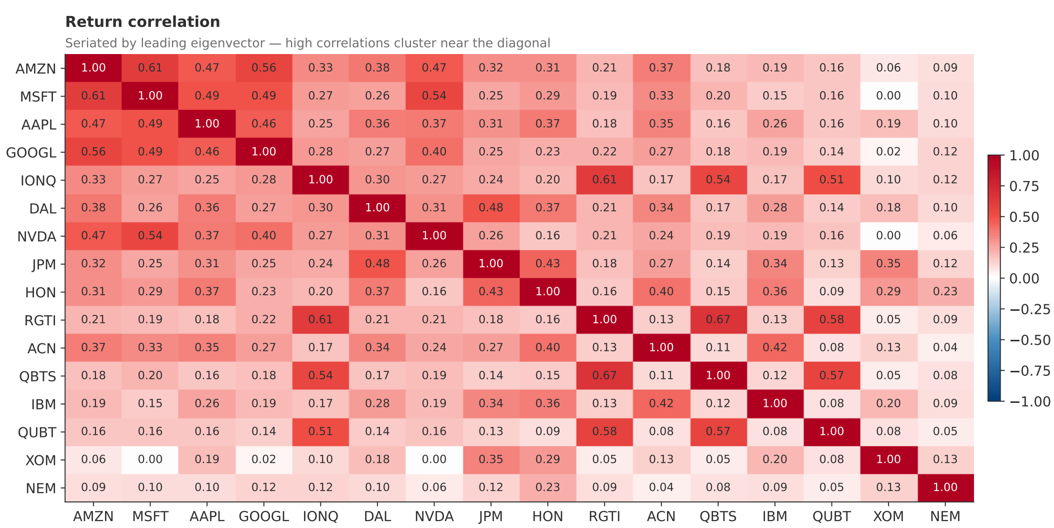

Figure 17. Pearson return-correlation matrix $\boldsymbol{\Sigma}_{ij}/\sqrt{\boldsymbol{\Sigma}_{ii}\boldsymbol{\Sigma}_{jj}}$ across all sixteen tickers, reordered by the leading eigenvector. The dense block corresponds to the mega-cap technology cluster, while the anti and small-cap quantum baskets form weaker co-movement blocks.

C. Motivational figures

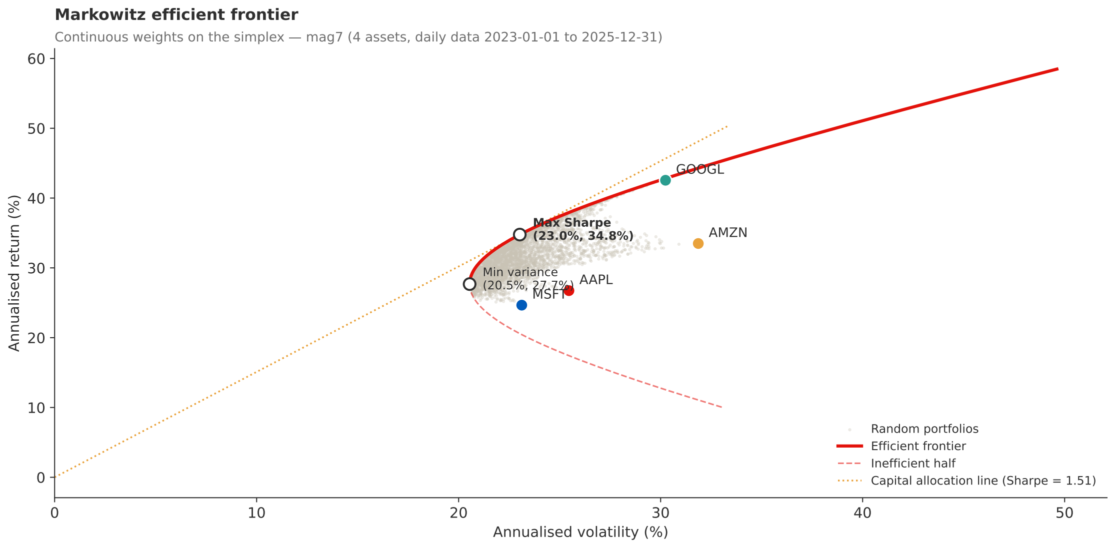

Figure 18. Continuous Markowitz efficient frontier on the mag7 sub-basket ($n = 4$: AAPL, MSFT, GOOGL, AMZN). Random simplex-sampled portfolios form the cloud, the analytical efficient frontier traces the feasible upper envelope, and the labelled markers give per-asset annualised risk and return.

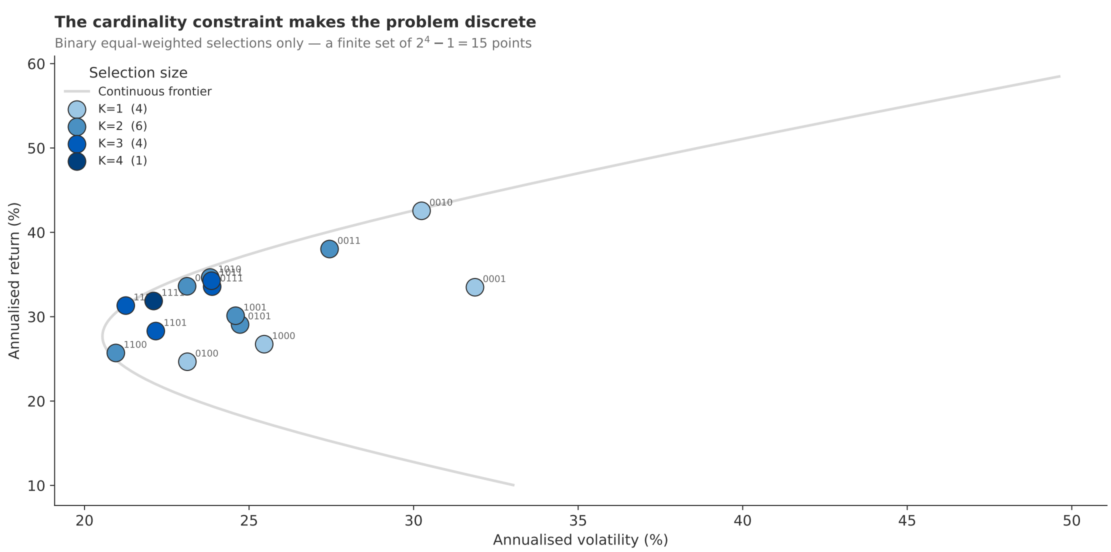

Figure 19. The cardinality constraint discretises the mag7 frontier ($n = 4$). Binary equal-weighted selections form a finite set of $2^n - 1 = 15$ points, with only the $\binom{n}{K}$ points at the prescribed budget admissible for the cardinality-constrained problem.

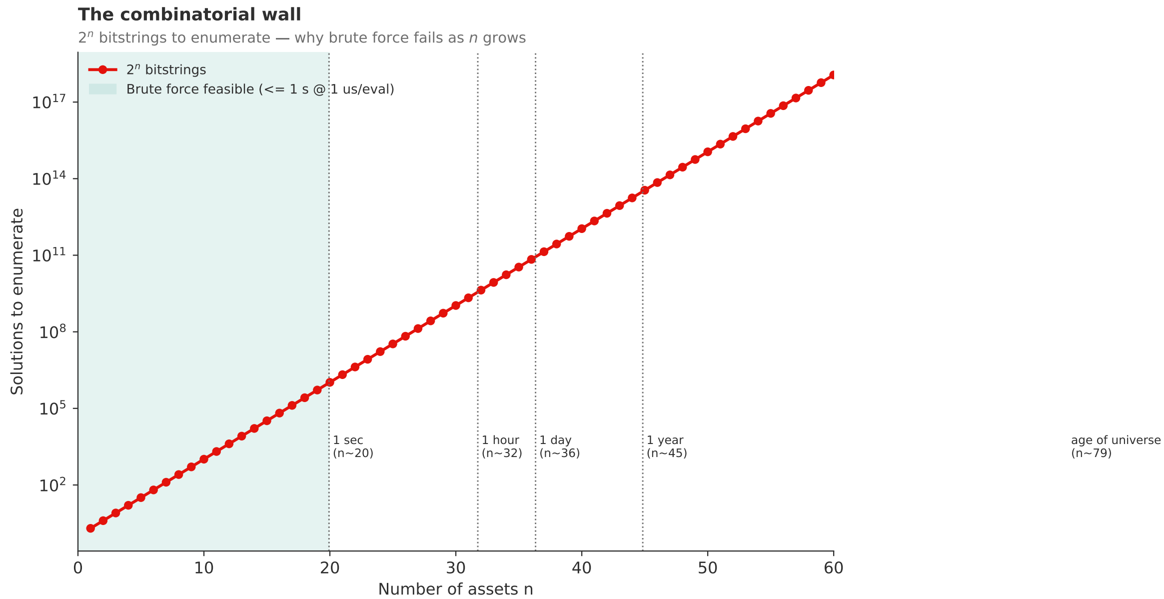

Figure 20. The combinatorial wall on the canonical mag7-extended instance: 2^n bitstrings on a log vertical axis as n grows, with annotated wall-clock budgets at one microsecond per evaluation. At $n = 16$ the brute force is still tractable, motivating the use of a heuristic such as QAOA for larger regimes.

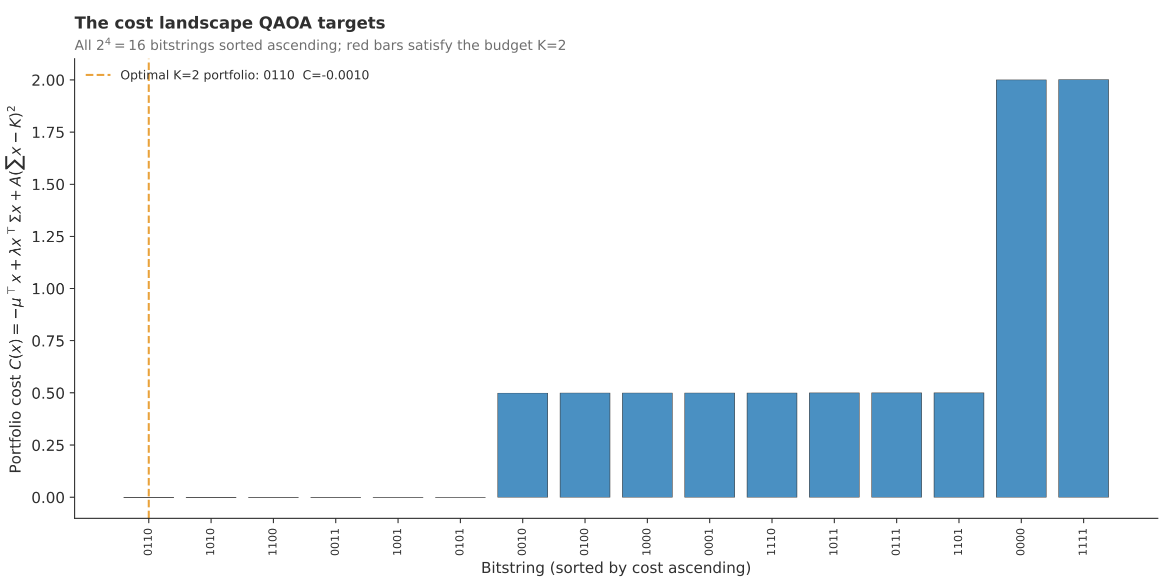

Figure 21. Cost landscape on the canonical instance ($n = 16$, $K = 4$, $\lambda = 2.0$, $A = 0.5$): all $2^{16} = 65\,536$ bitstring costs sorted in increasing order. Budget-feasible states with $\sum_i x_i = K$ are highlighted in red; the feasible bitstrings are scattered through the spectrum rather than clustered.